



Mixed-Signal Electrical Interfaces

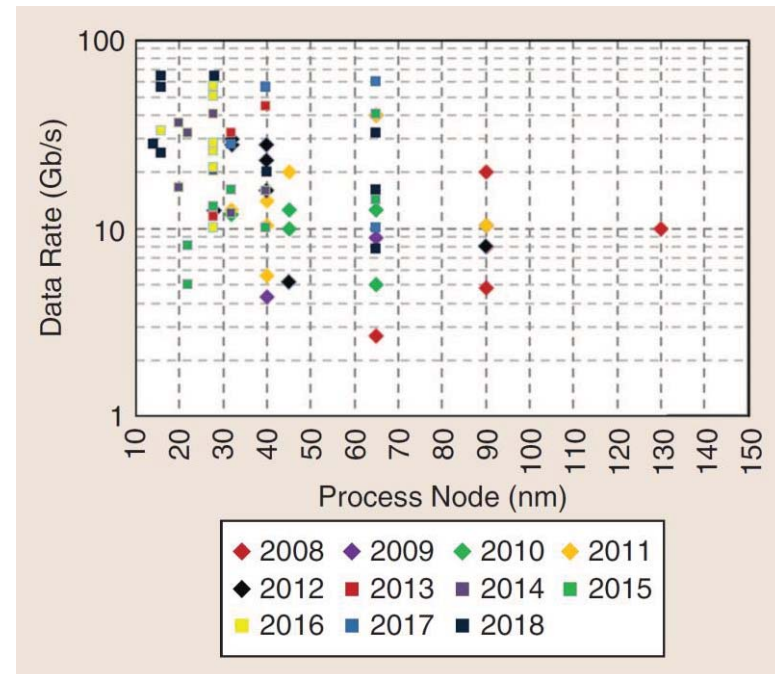
Elad Alon



University of California, Berkeley

The SerDes Problem in a Nutshell

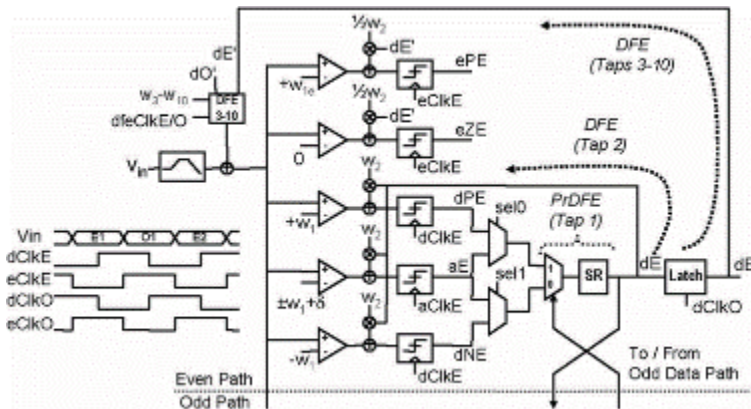
- Required throughput doubles every ~4 years
 - 56G/112G per lane rolling out now
- Power budget ~constant
 - Energy-efficiency (pJ/bit) must improve
 - But system vendors would love to not have to upgrade their channels
 - And process may or may not help much any more



**from ISSCC Tech Trends 2018*

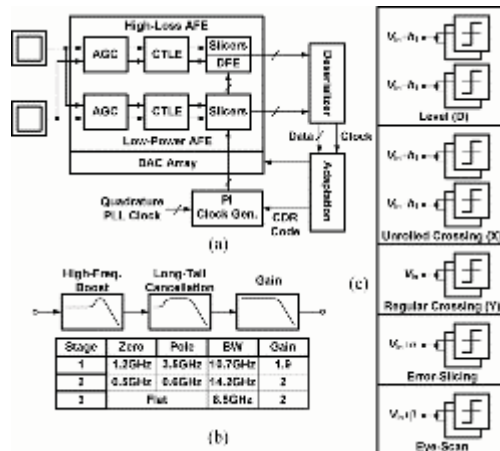
SerDes “Golden” Architecture (2005 - 2018+)

Rambus



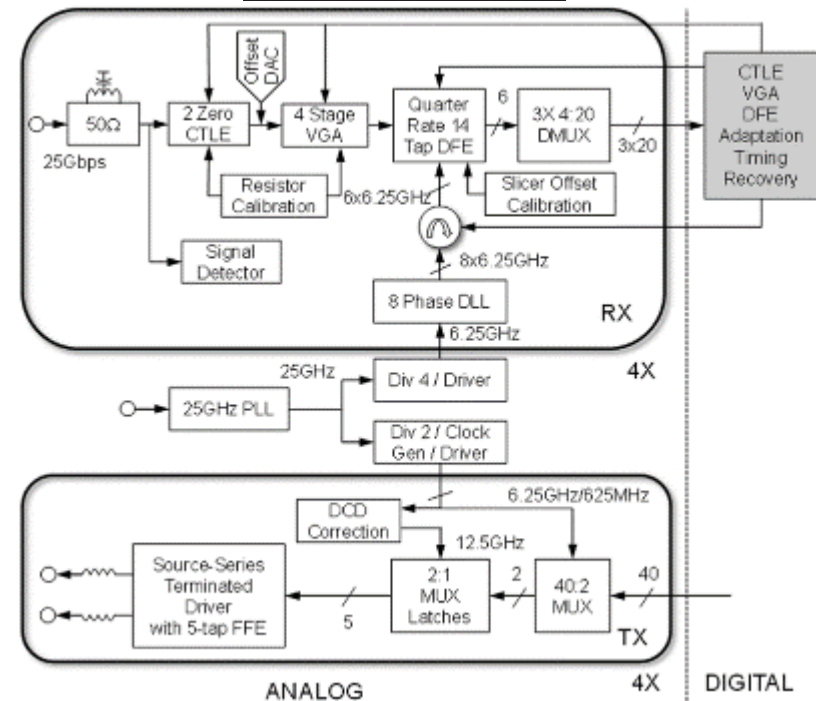
B. Leibowitz et al., ISSCC 2007

Xilinx



J. Savoj et al., VLSI 2012

Broadcom

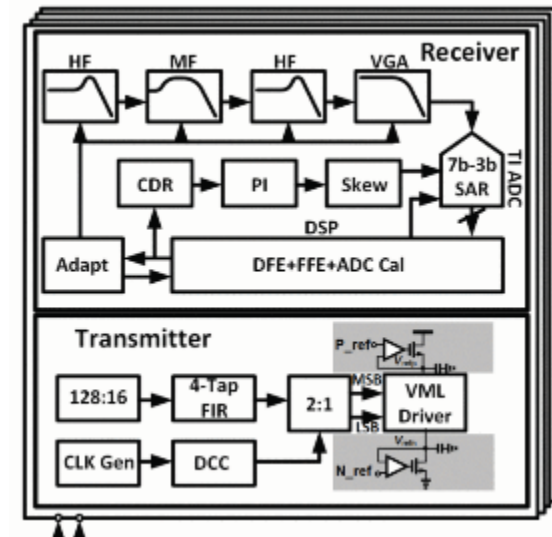


B. Zhang et al., JSSC 2015

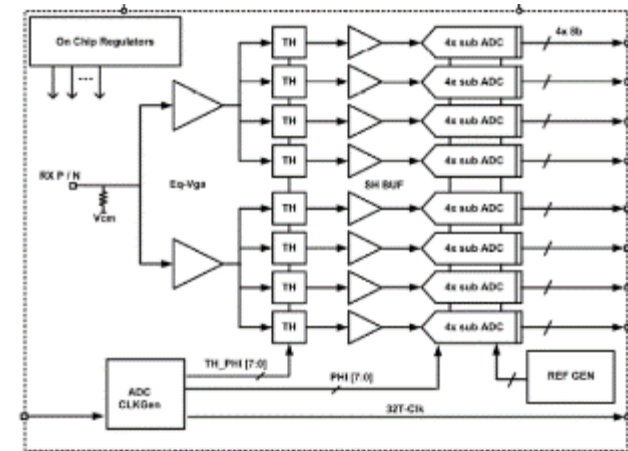
- NRZ
- TX FFE, CTLE, DFE
- LMS-based adaptive equalization
- 2X oversampled CDR

Didn't I Just Hear a Great Talk About ADC-Based Serdes?

- ADC-based designs have indeed gained significant traction in the 56G/112G generation
- This talk aims to provide insights in to:
 - Why analog/mixed-signal was preferred
 - Why one might choose ADC+digital vs. analog (or a mix)
 - Key techniques for mixed-signal implementations



P. Upadhyaya, ISSCC 2018



K. Gopalakrishnan, ISSCC 2016

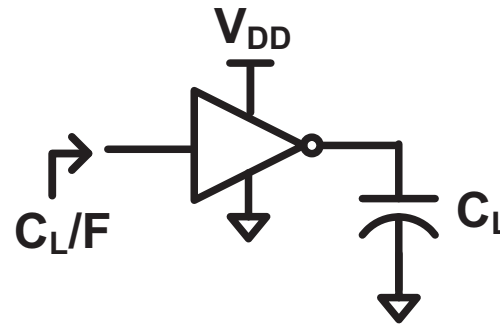
Outline

- Introduction
- **Power in SerDes Circuits**
- **Implications on SerDes Circuits and Architectures**
- **60Gb/s 65nm Design Example**
- **Conclusions**

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Component #1: Digital Power



- Neglecting leakage and crowbar:

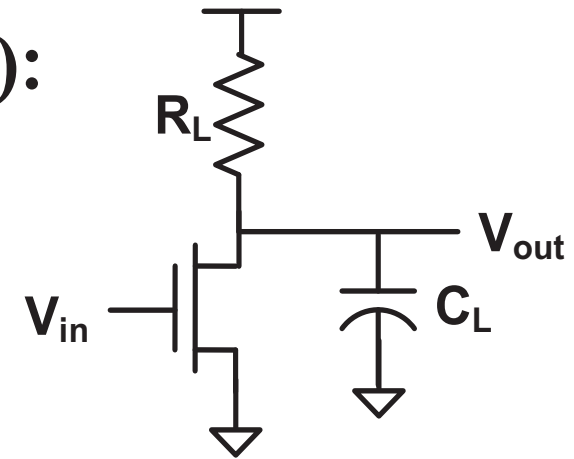
$$P_{dig} = \alpha_{0 \rightarrow 1} C_L V_{DD}^2 f_{clk} \frac{1 - t_{inv}/t_{del}}{1 - \gamma_d t_{inv}/t_{del}}$$

- V_{DD} fixed by throughput (delay) constraints
 - So C_L is largely what sets the power
 - (Don't try to go too fast relative to the process capability)

Component #2: Analog Power (Classic)

- When limited by SNR (dynamic range):

$$P_{ana} \approx \gamma_n SNR_{\min} k_b T \omega_{bw} \frac{V_{DD} V_{i,\max}}{V_{sig}^2}$$

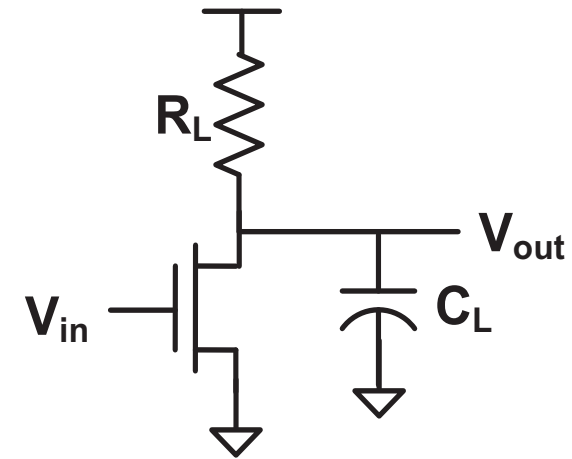


- Name of the game is maximizing signal relative to supply/total input swing
 - Explicitly add load capacitance to meet required SNR
 - 1 extra bit = 4X SNR = 4X power...
- What if minimum load capacitance gives better SNR than needed?

GBW-Limited Analog Power

- Gain-bandwidth limited:

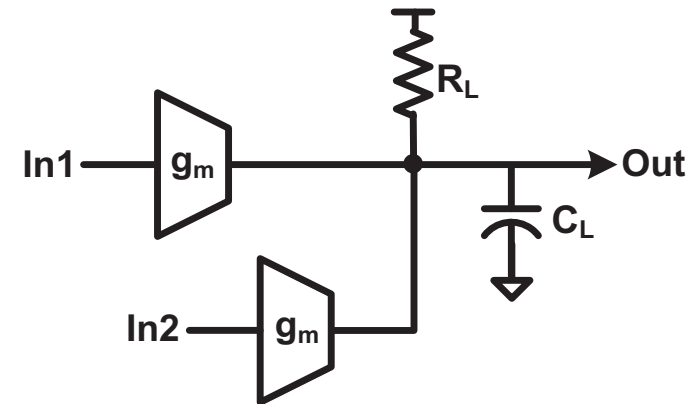
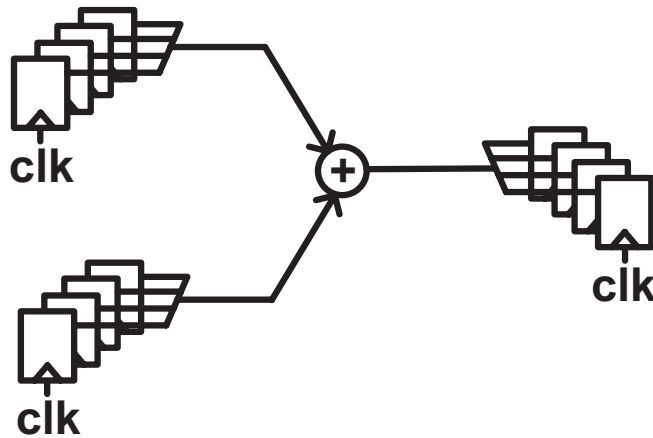
$$P_{ana} = C_L V^* V_{DD} A_V \omega_{bw} \frac{1}{1 - \frac{A_V \omega_{bw}}{\omega_T / \gamma_d}}$$



- Looks/behaves a lot like digital power
 - Gain, bandwidth usually decided at system level (based on throughput/delay)
 - So again, C_L is often the main factor setting the power
 - (Still, don't try to go too fast relative to process)

Key Implication

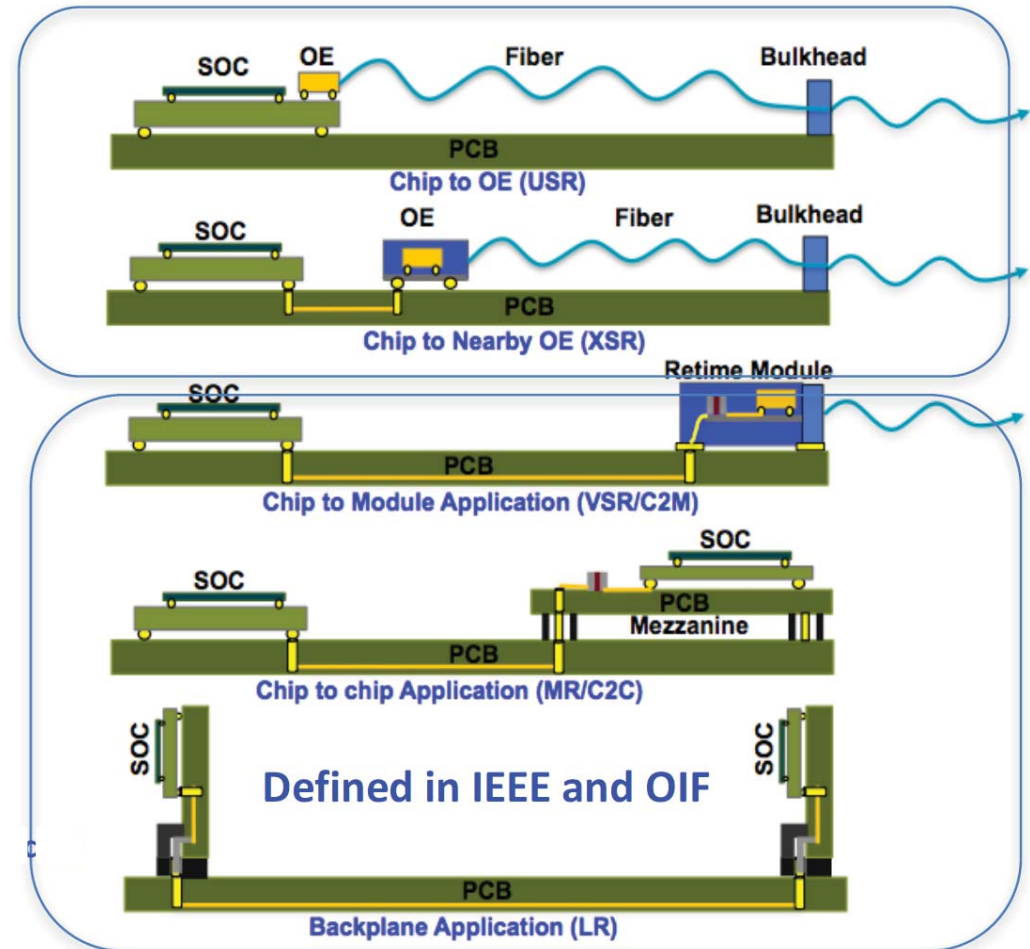
- High dynamic range clearly favors digital
 - Adding 1 more bit vs. 4X power



- For low dynamic range (~5-6 bits), it's all about the total capacitance
 - C_{total} for analog can be substantially smaller than digital in this regime, and “Golden Age” SerDes were definitely operating here

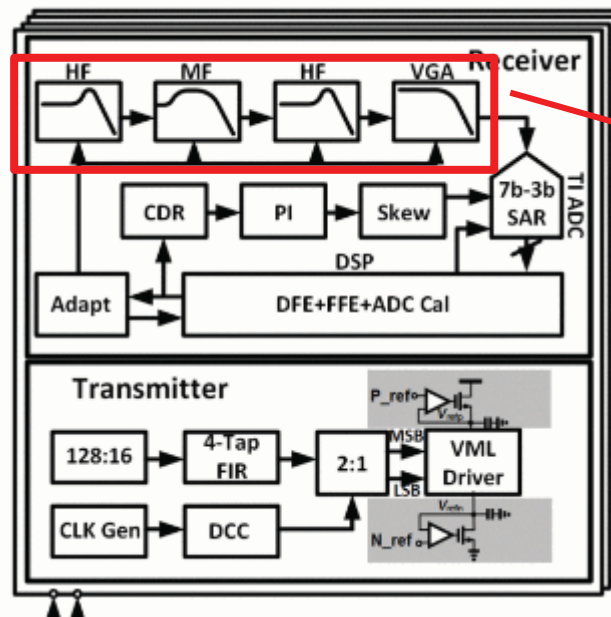
So Did Things Change (In Terms of DR)?

- In some applications, maybe
 - LR, lossy (but still smooth) channels could start pushing front-end DR above 5-6 bits
- However, almost all ADC+DSP designs still have analog pre-processing...

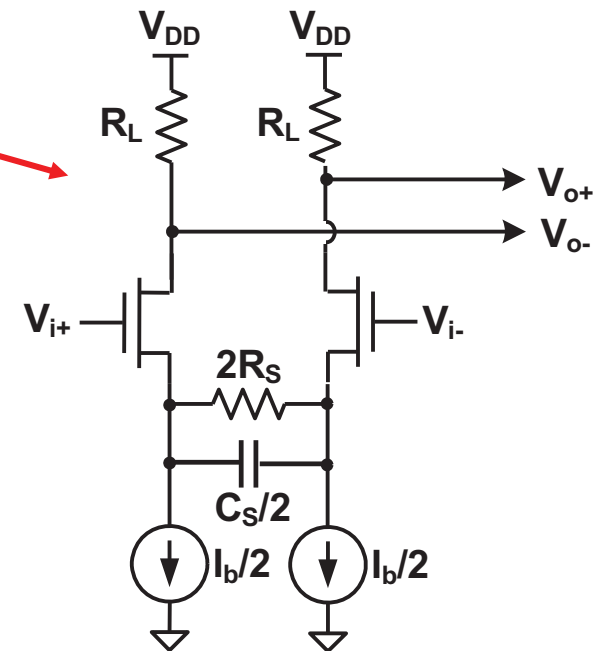


A. Ghiassi, *IEEE 802 meeting*, Jan. 2018

Analog Pre-Processing Example: CTLE



P. Upadhyaya, ISSCC 2018



- For $V_{DD} = 0.8V$, $A_{pk} = 2$, $\omega_{bw} = 60\text{Grps}$ ($\sim 28\text{GS/s}$), $C_L = 10\text{fF}$, and typical $\sim 16\text{nm}$ process

$$P_{ana} = 274\mu\text{W}$$

Analog CTLE vs. Digital IIR Filter

- At 0.8V and with same net throughput (28GS/s), can afford at most $C_{\text{eff}} (= \alpha_{0 \rightarrow 1} C_L)$ of 15fF for same power as analog CTLE
 - (This is also not counting extra ADC resolution to maintain linearity)

Analog CTLE vs. Digital IIR Filter

- At 0.8V and with same net throughput (28GS/s), can afford at most C_{eff} ($=\alpha_{0\rightarrow 1}C_L$) of 15fF for same power as analog CTLE
 - (This is also not counting extra ADC resolution to maintain linearity)
- Typical flip-flop $\sim 2\text{-}3\text{fF}$ of C_{eff} , so would need to implement the **IIR with $< \sim 5\text{-}8$ flip-flops** to win over analog
 - Parallelism and lower V_{DD} can improve digital, but gains somewhat limited due to IIR feedback loop
 - This is why ADC-based SerDes typically still have analog CTLEs

Outline

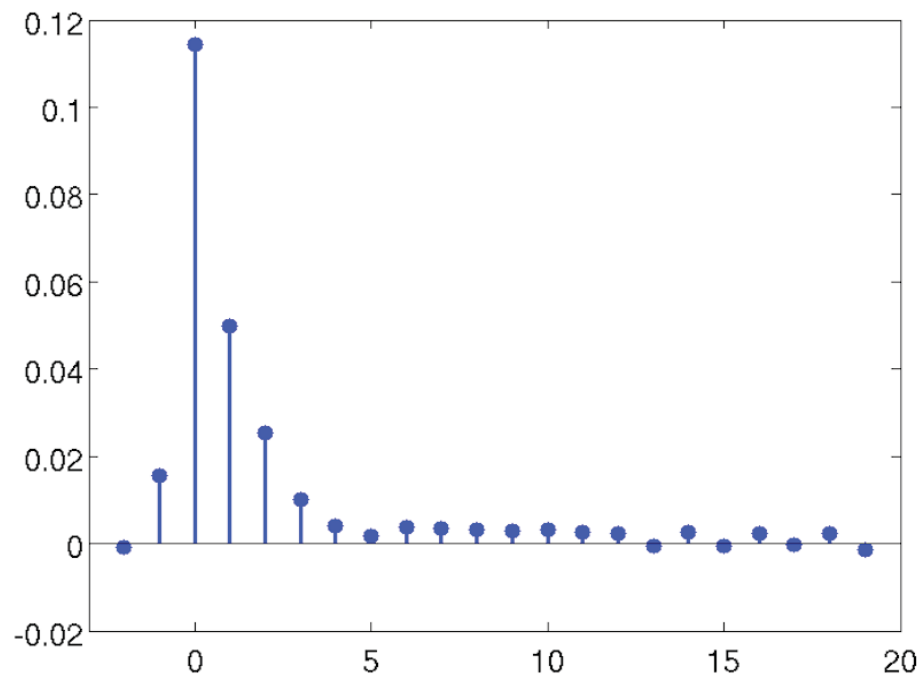
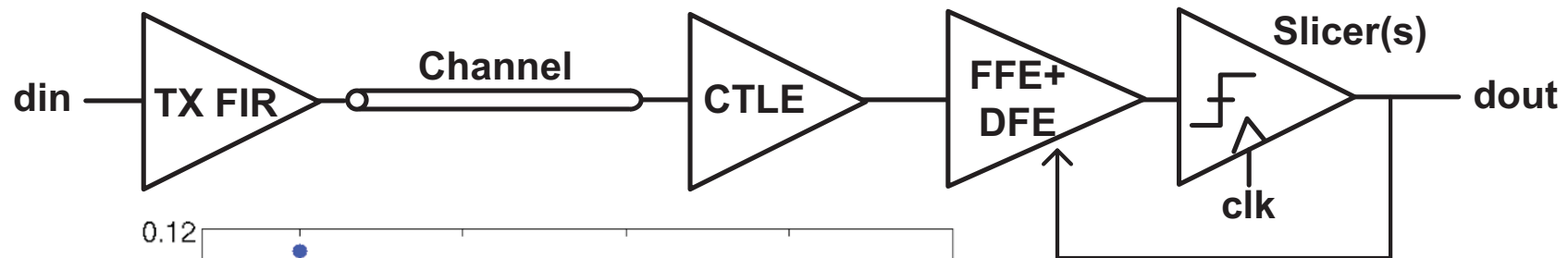
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- Power in SerDes Circuits
- **Implications on SerDes Architectures**
 - Equalization Architecture and Circuits
 - Clocking and CDR Architecture
- 60Gb/s 65nm Design Example
- Conclusions

Important Note

- Equalization architecture discussion will mostly focus on links without forward error-correction (FEC)
 - And will also mostly show NRZ examples
- **However, almost all of the techniques/concepts remain valid for PAM4 with FEC**
 - Main difference is that FEC will move us away from (multi-tap) DFE to longer FFEs

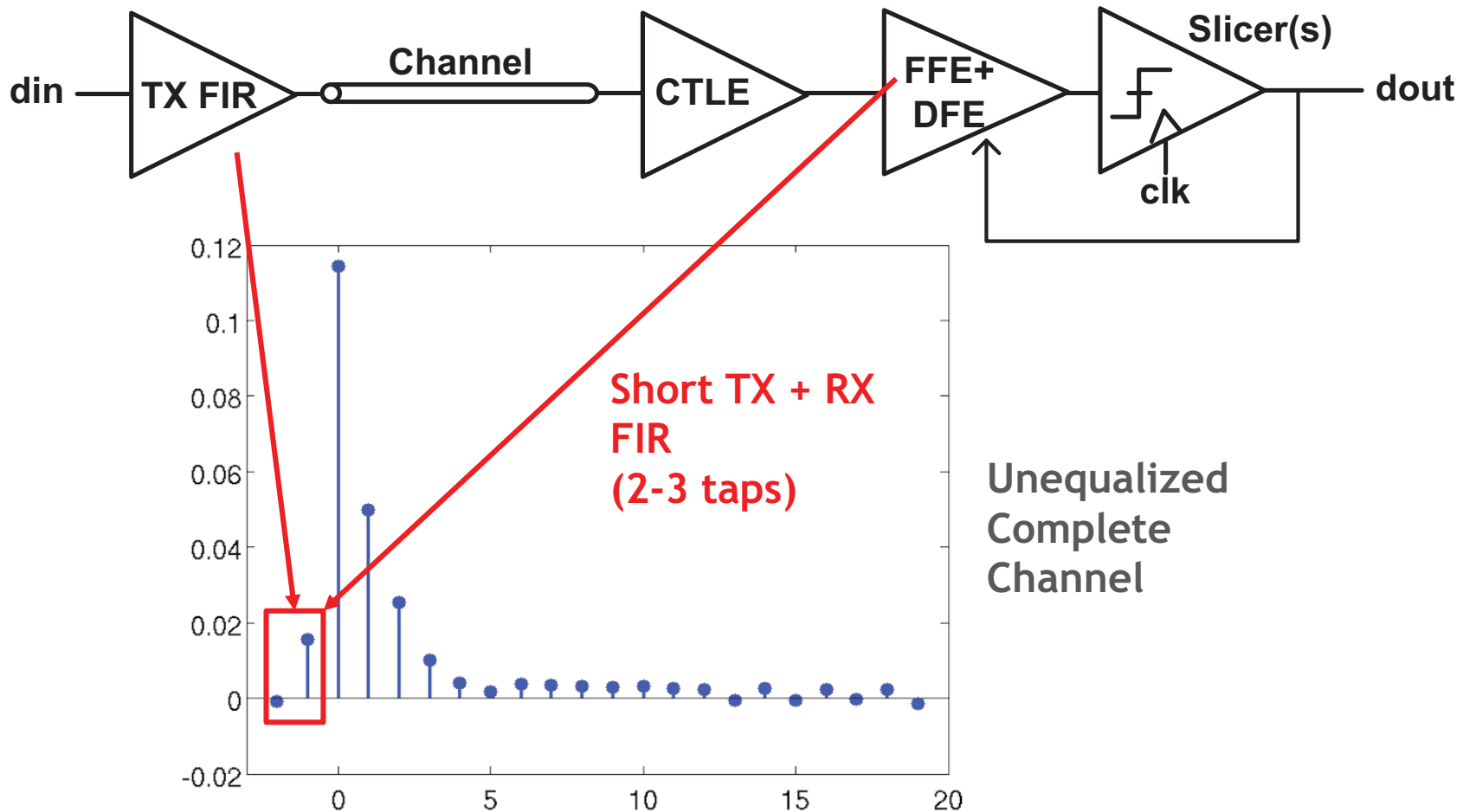
Equalization Architecture

- Even if “all analog”, efficient drives careful selection of what components handle what



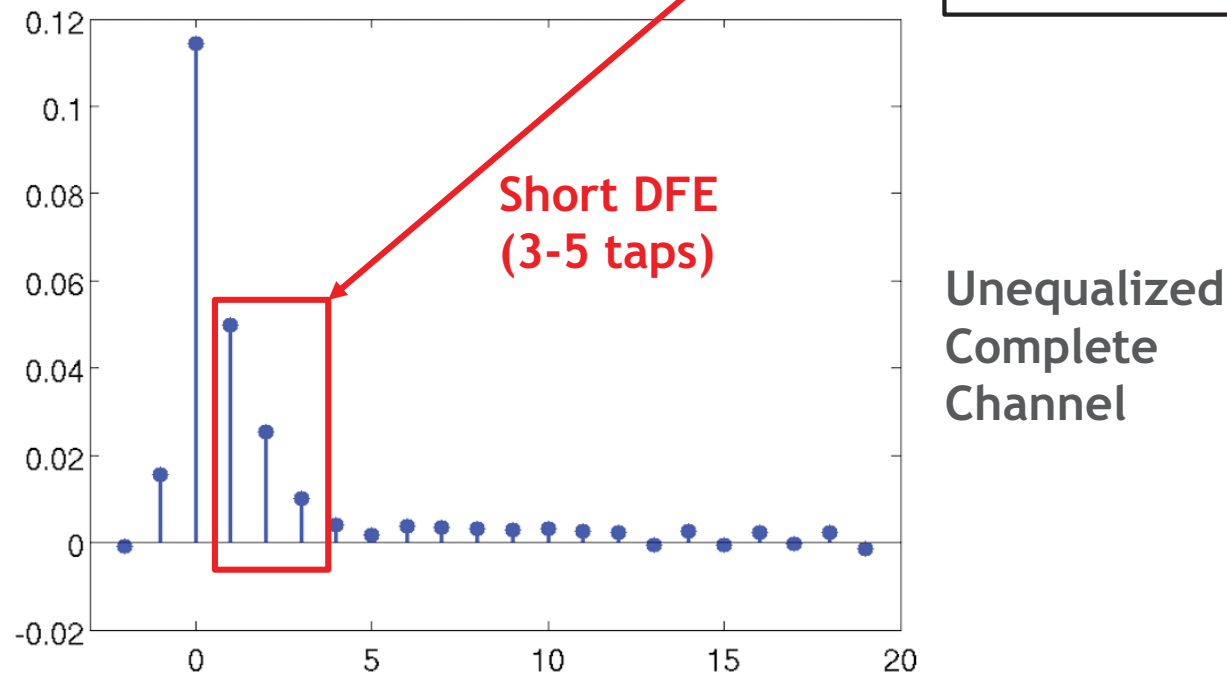
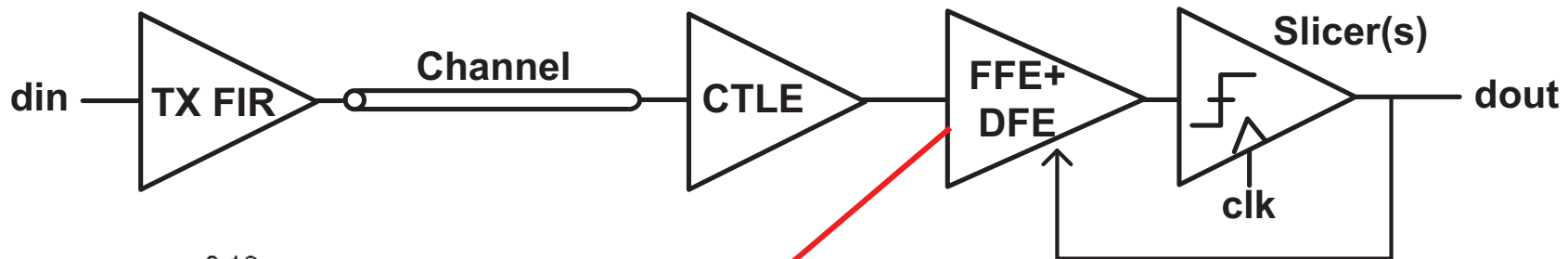
**Unequalized
Complete
Channel**

Equalization Architecture (2)



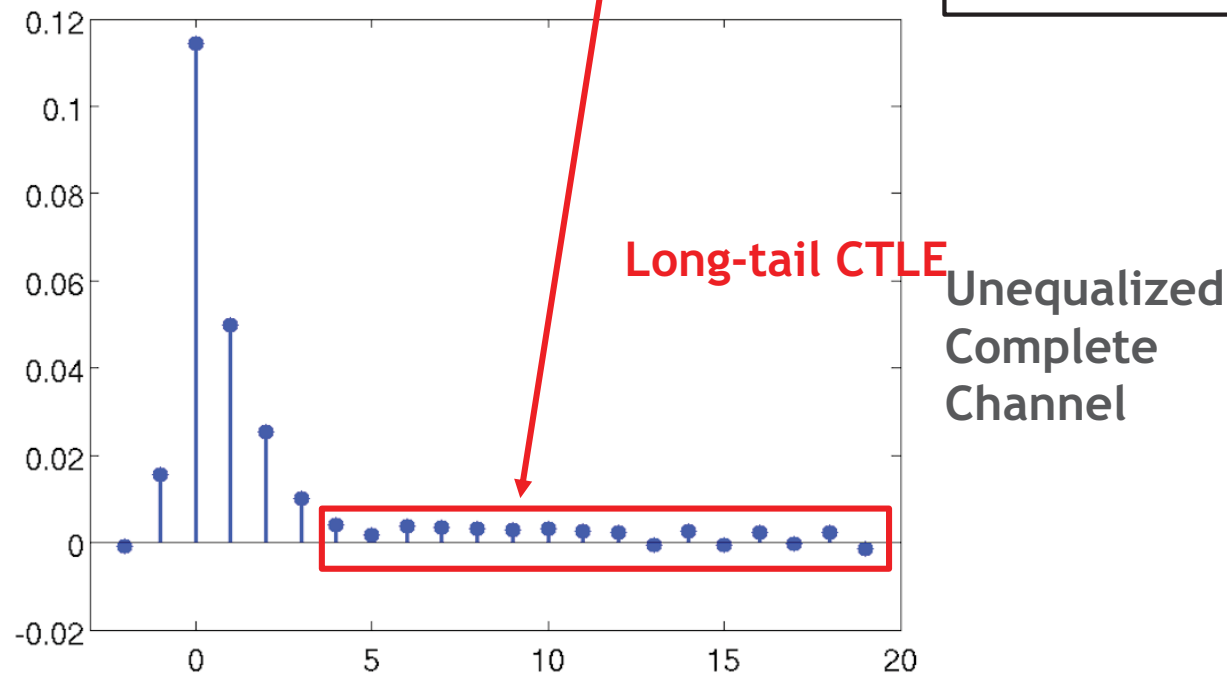
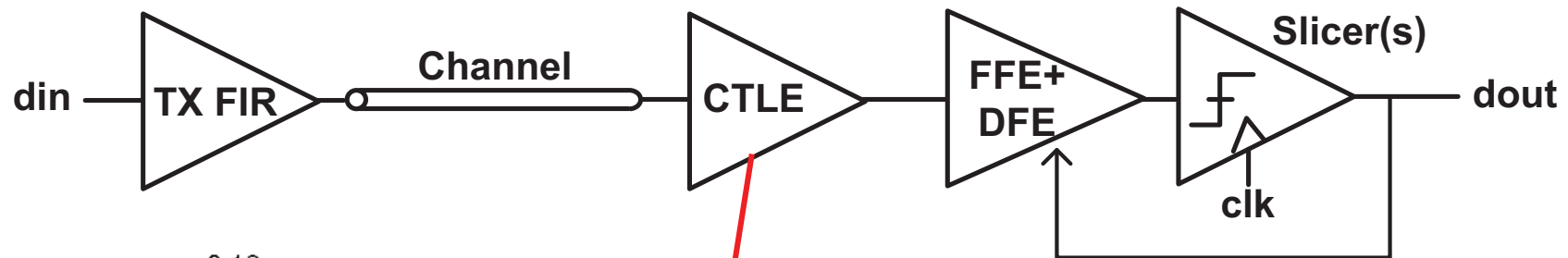
- Precursors can't be handled by DFE, hard to mimic with CTLE

Equalization Architecture (3)



- Keep DFE short to reduce power from delay elements (flops)

Equalization Architecture (4)



- CTLE area/power-efficient in realizing large time constants

Key Challenges at 56/112G

- **Efficiently realizing required gain-bandwidth**
- **Closing the (multi-tap) DFE feedback loop**
- **Realizing efficient analog FFEs**

Key Challenges at 56/112G

- **Signal chain typically includes ~4-8 stages of amplifiers before the slicers**
- **Even with DFE, required overall bandwidth $\sim (1/3)f_{\text{symbol}}$**
 - So need ~40-60GHz bandwidth per stage
 - With simple R-loaded stages, might start hitting technology limits (and hence burn very high power)

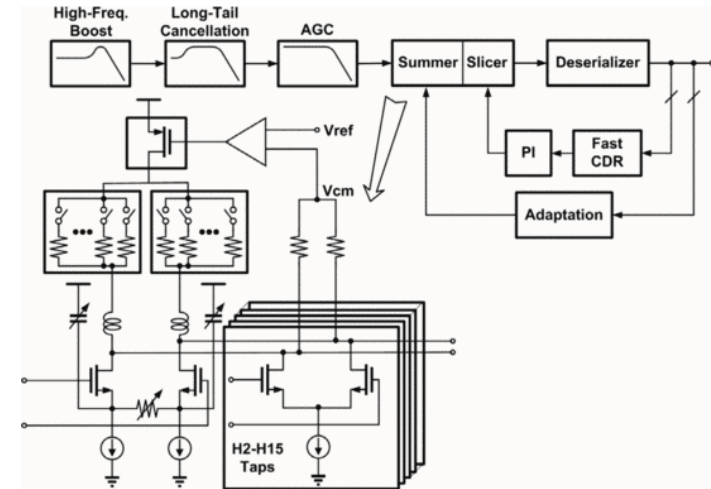
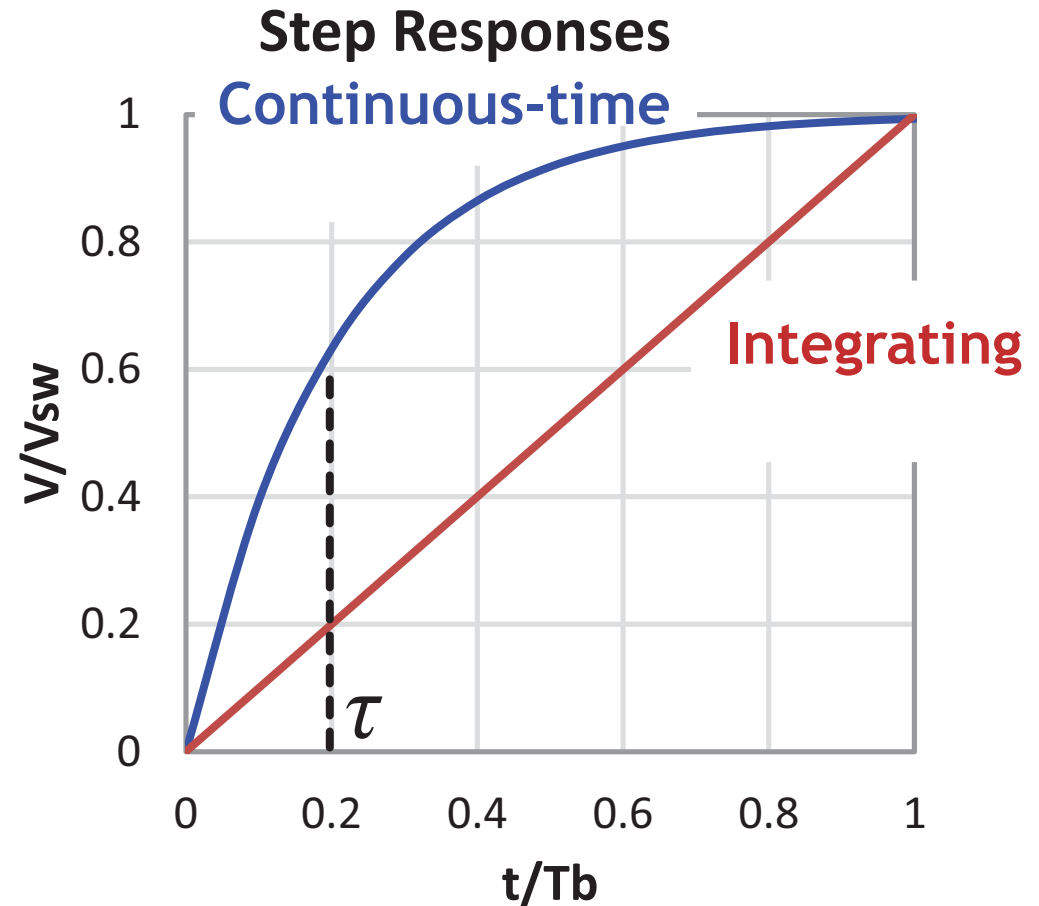
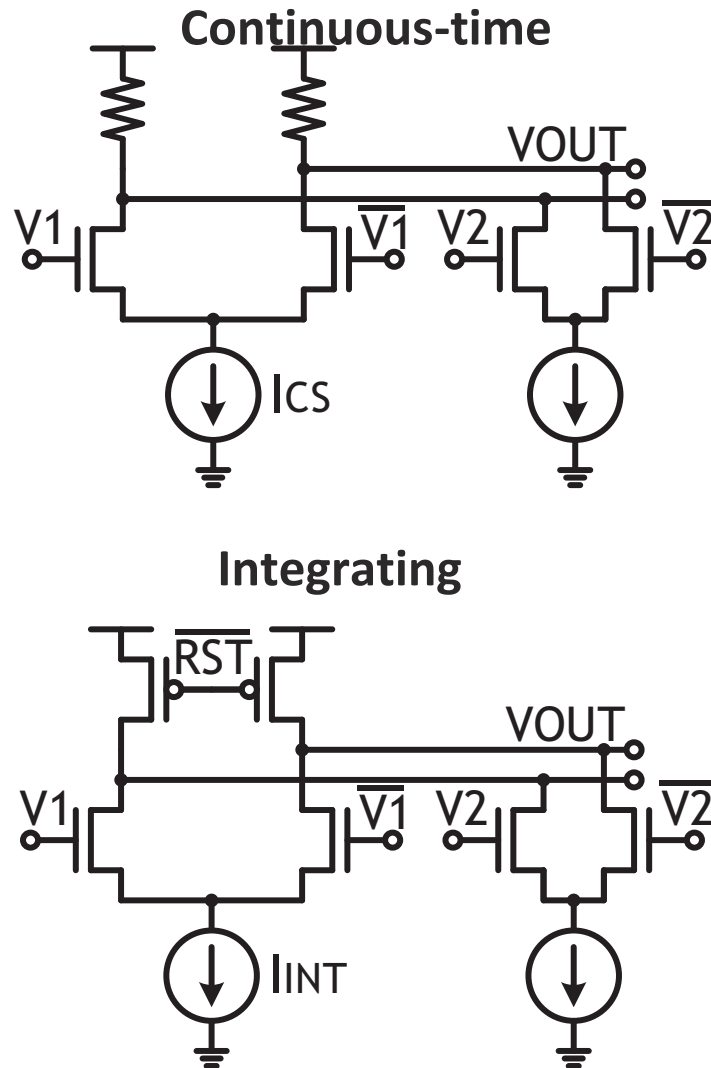


Figure from P. Upadhyaya, ISSCC 2015

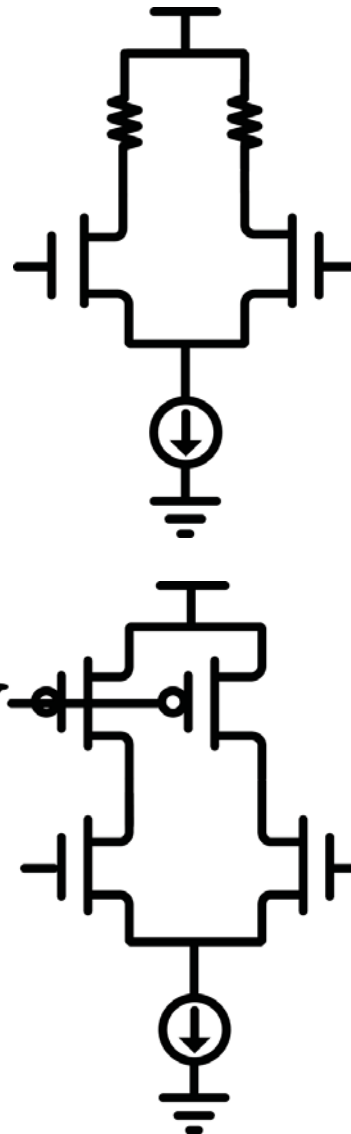
Improving Efficiency: Current Integration



M. Park, ISSCC 2007

Current Integration Benefits In Detail

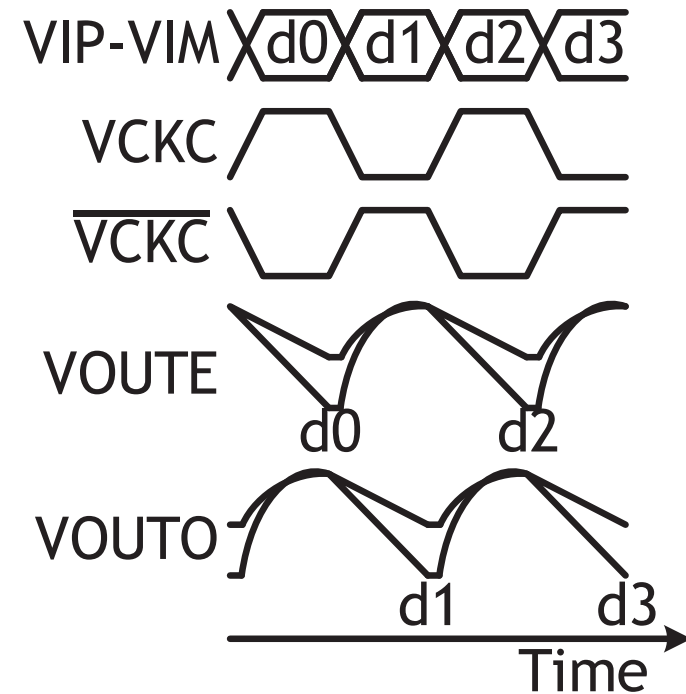
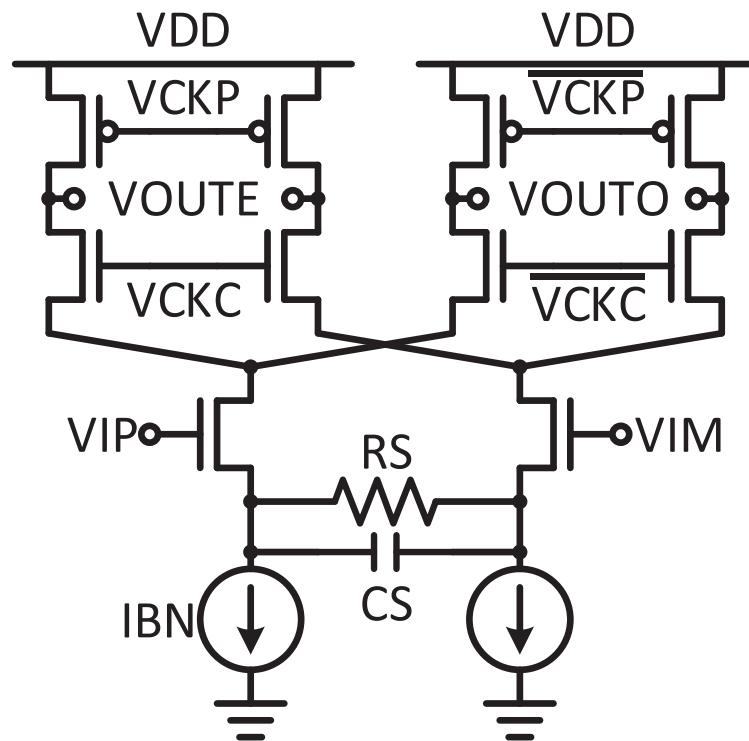
- Improves nominal current draw by N_τ (~3X)
- Also improves self-loading term by same factor
 - Which can be even more important at these rates



$$I_{cont_time} = \frac{A_v V^* C_L N_\tau / T_{bit}}{1 - \frac{A_v N_\tau / T_{bit}}{(\omega_T / \gamma)}}$$

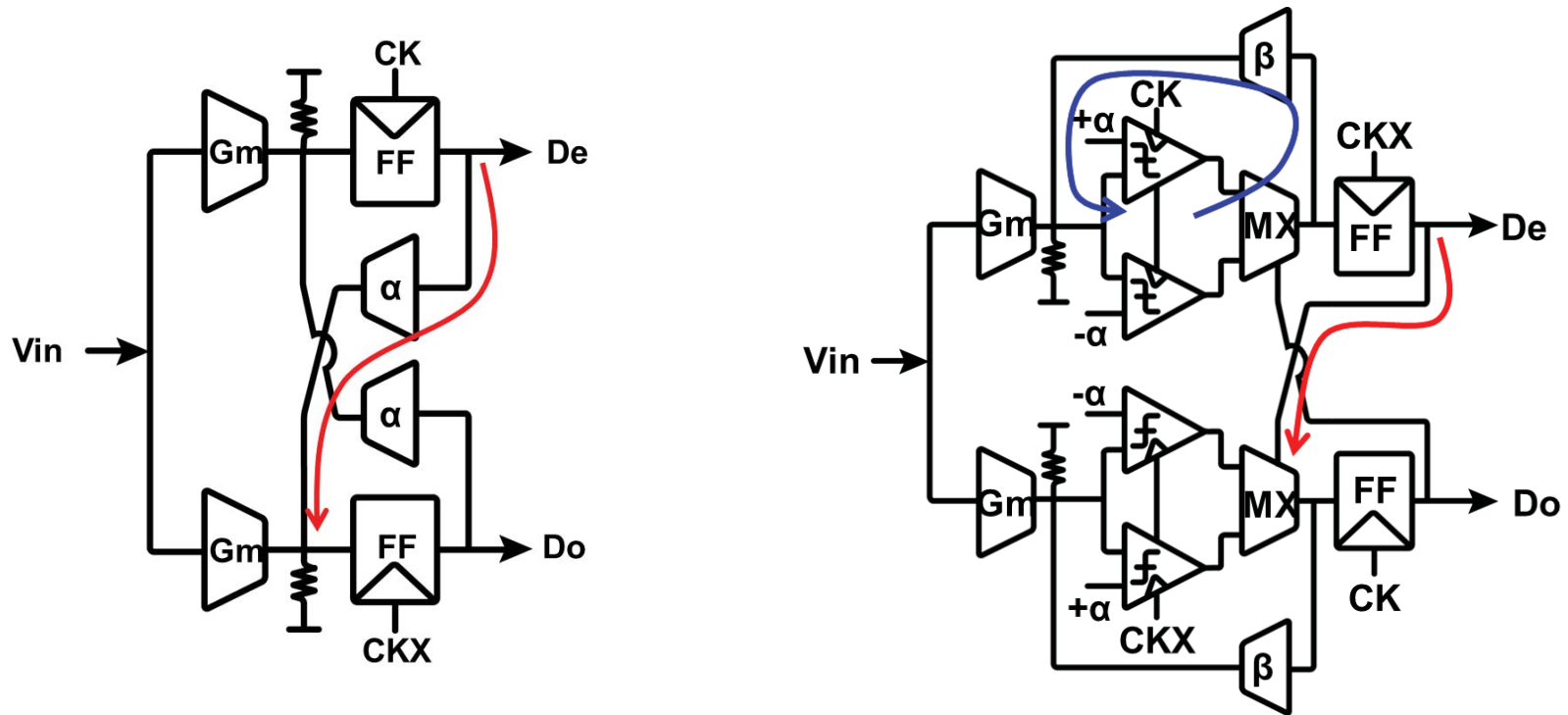
$$I_{current_int} = \frac{A_v V^* C_L / T_{bit}}{1 - \frac{A_v / T_{bit}}{(\omega_T / \gamma)}}$$

Ex: Current Integrating CTLE/2:1 DeMUX



- Integrating the input may be advantageous for other reasons too (more soon)

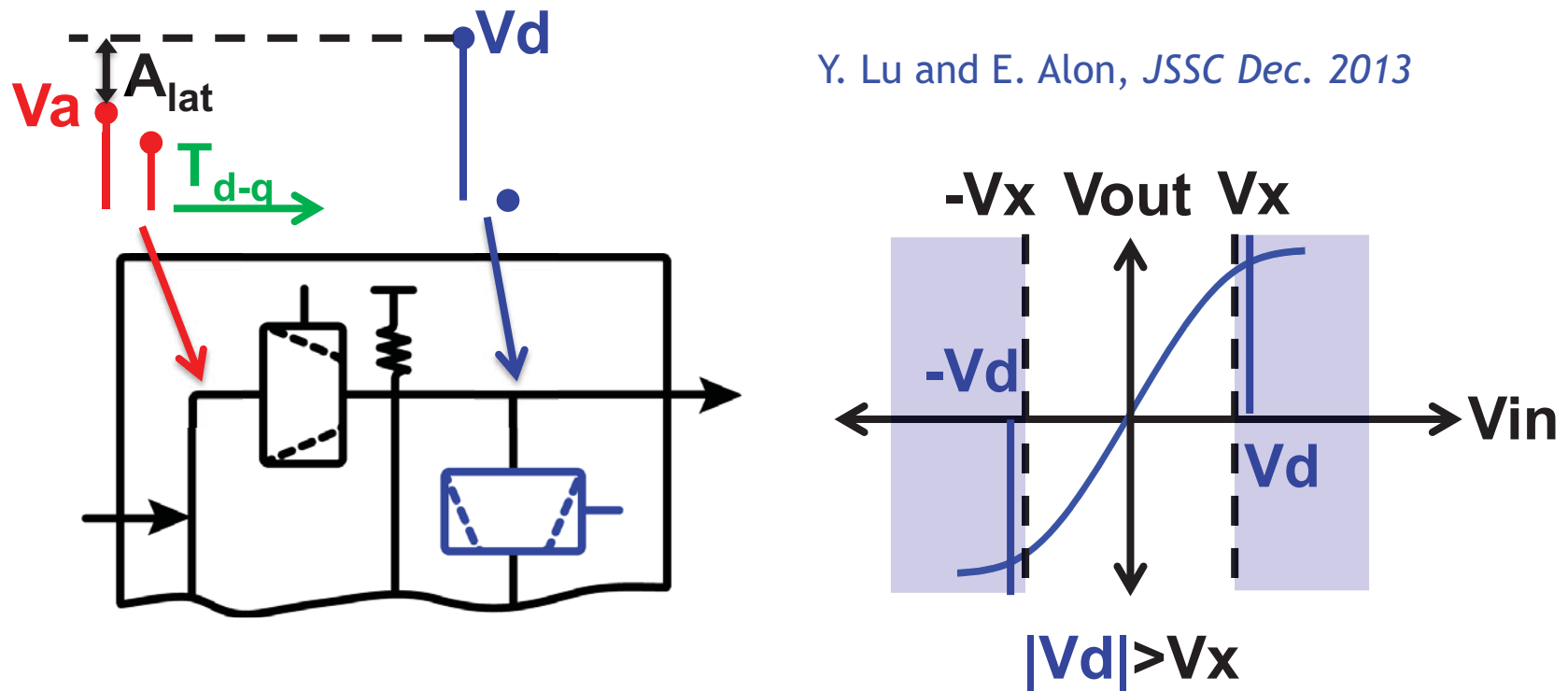
DFE 1st Tap Timing



- **Loop unrolling most common approach today**
- **But sacrifices timing of later taps**
 - (And requires a lot of samplers/hardware in PAM4 designs)

Key Idea: Pull DFE Gain Out of the Loop

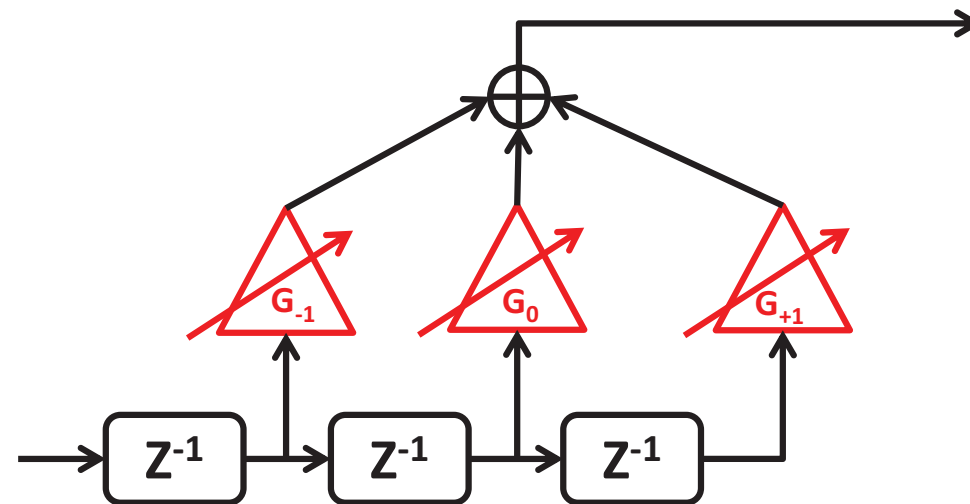
Y. Lu and E. Alon, *JSSC Dec. 2013*



- **DFE latch doesn't actually need high gain**
 - Just need enough to drive the feedback devices into the non-linear regime
- **Typically need latch gain of only ~ 2**

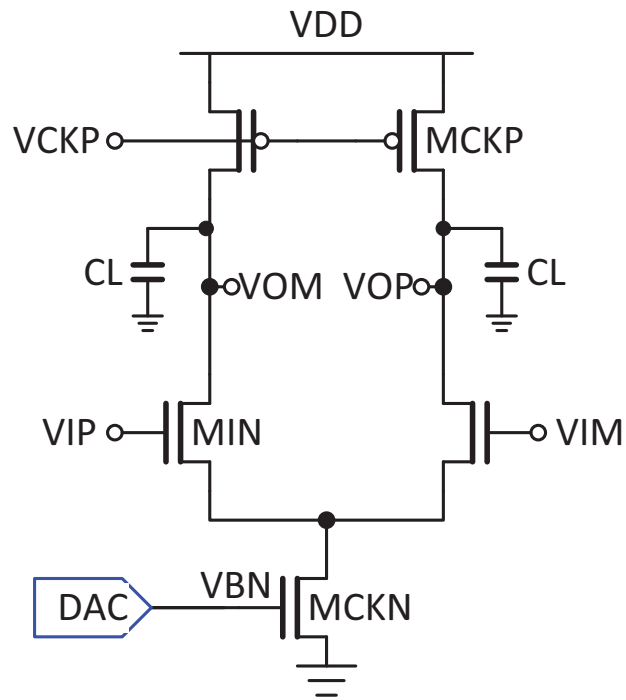
Mixed-Signal RX FFE

- **Focus first on relatively short FFEs**
 - I.e., where we don't need to worry about accumulating errors from analog delay elements

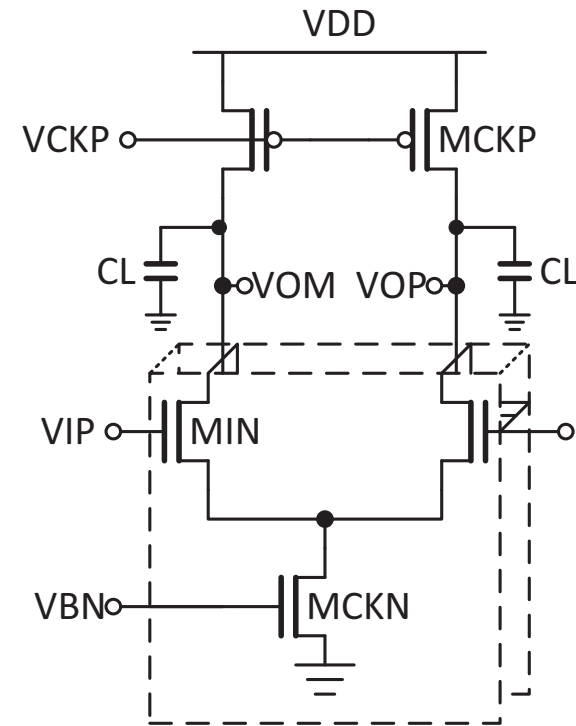


- **In this regime, key challenge is implementing variable gain efficiently**

Common VGA Designs

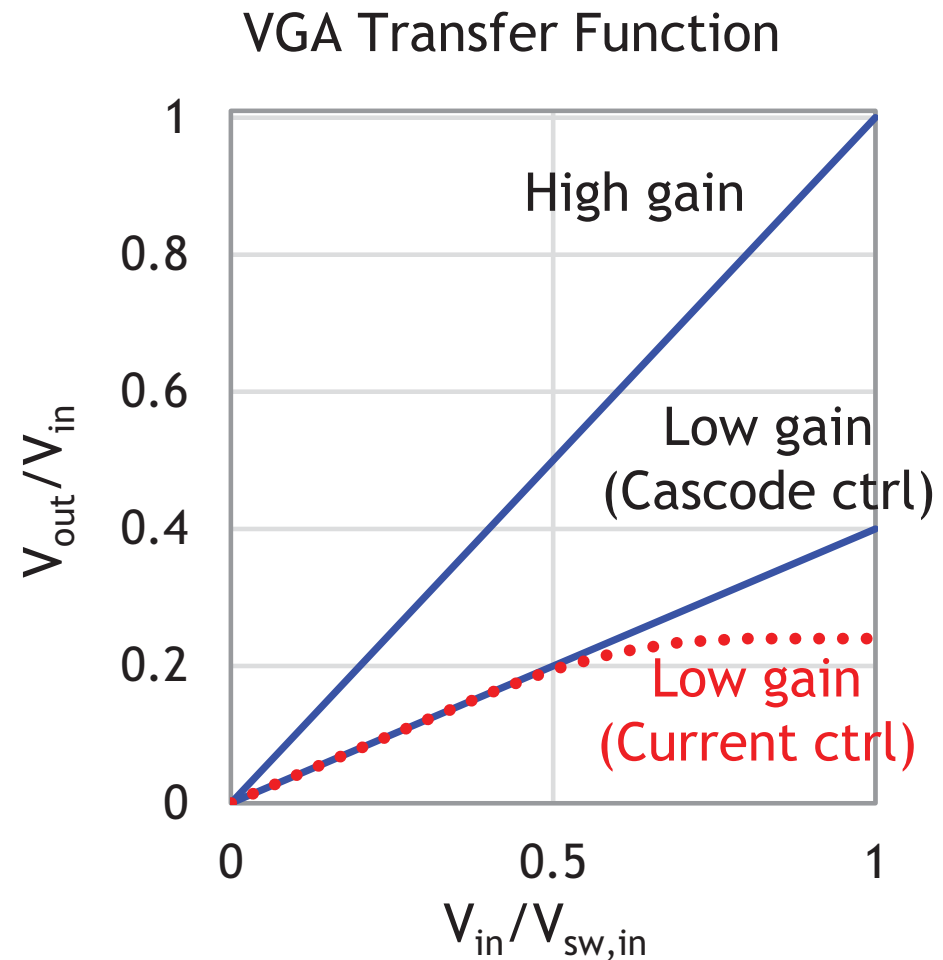
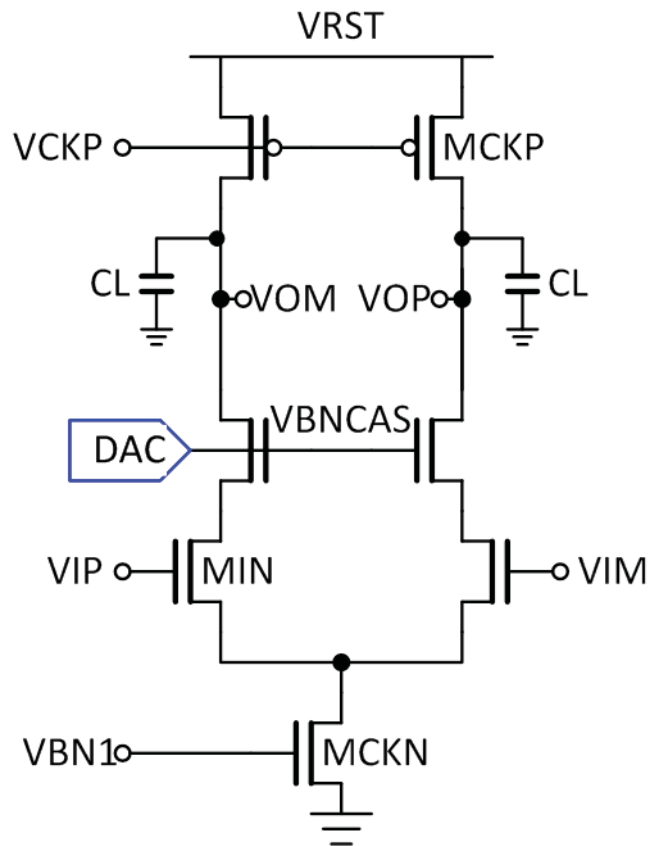


- **Variable I_{tail} : poor linearity @ low gain**



- **High unit cell parasitic capacitance**
 - Keep dynamic range off of the signal path...

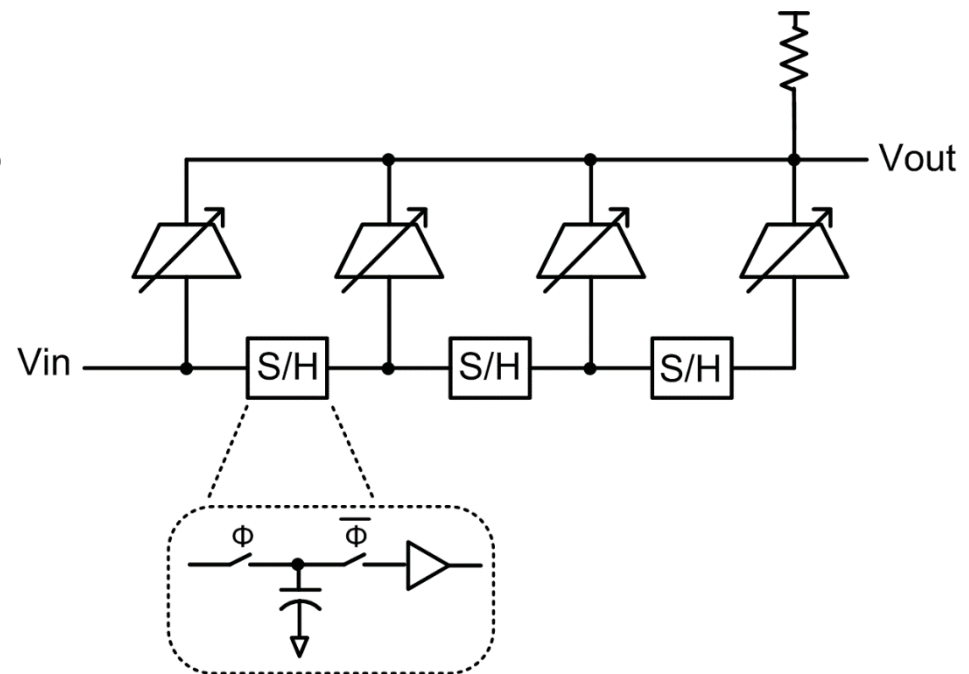
Solution: Variable Bias Cascode



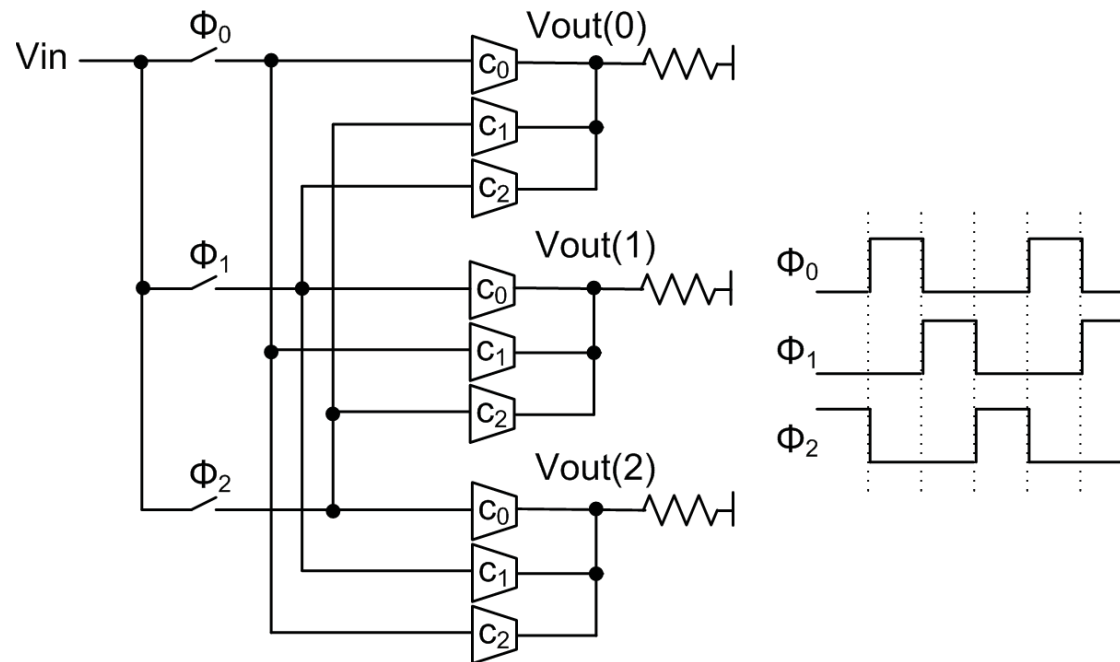
- **Minimum linearity degradation over gain range**

What if We Actually Want a Long FFE?

- **Key issue with long analog FFEs is the delay line**
 - I.e., where we don't need to worry about accumulating errors from analog delay elements
- **Every sampling stage adds at least kT/C noise**
 - So need $\sim 10\text{-}20\text{fF}$ per stage for 10-20 taps
- **Deterministic errors accumulate too...**
 - Watch out to not rail...



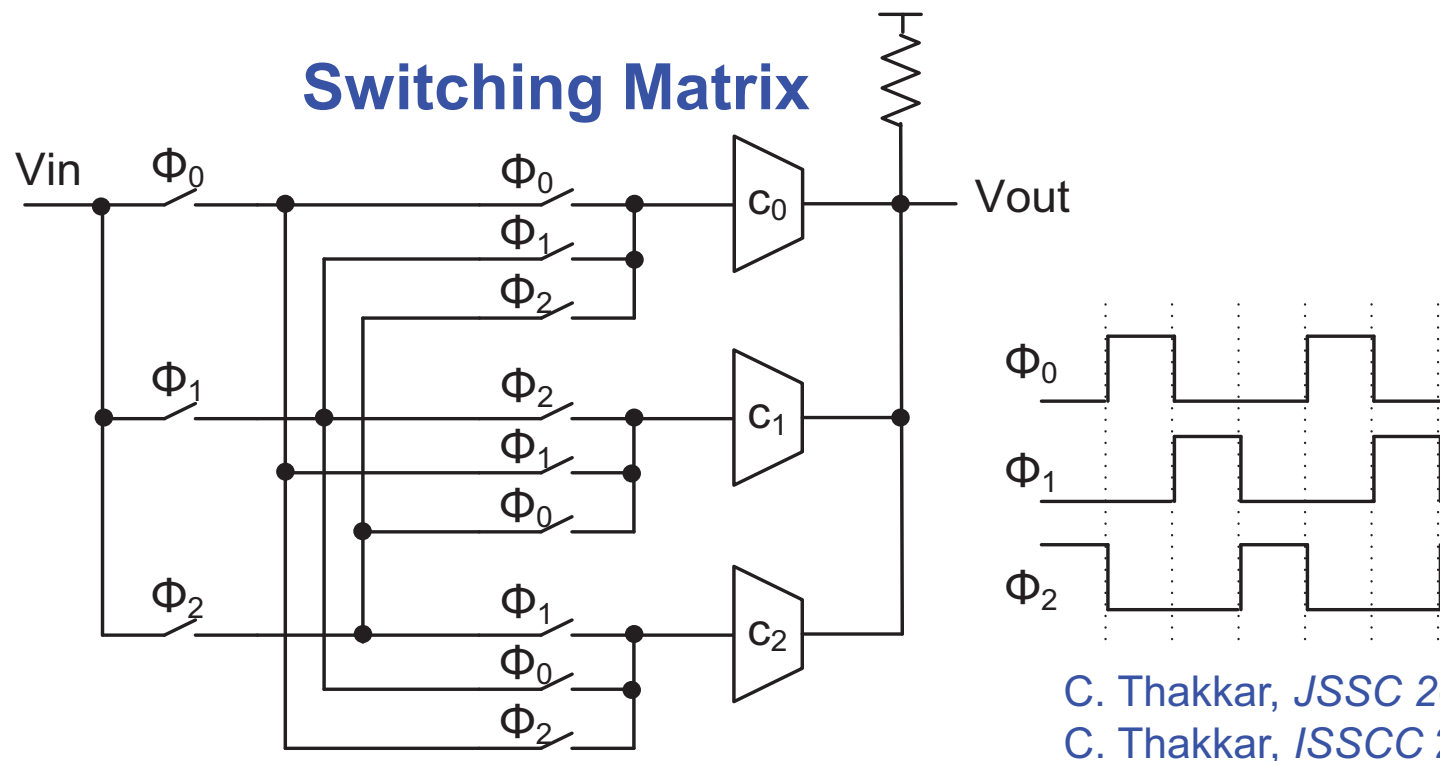
(Analog) Parallelism



- Only one S/H needed to generate the delay – no accumulation
 - But might become physically large / wire constrained with large number of taps

J. Jaussi, JSSC 2005

Switching Matrix Architecture



- **Switching matrix re-serializes the array of (delayed) outputs from 1st stage deserializer/sampler**
 - So that a single weight amplifier can be used per coeff.
 - Avoids wire capacitance overheads of fully parallel design

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CDR Architecture: Dual Loop?

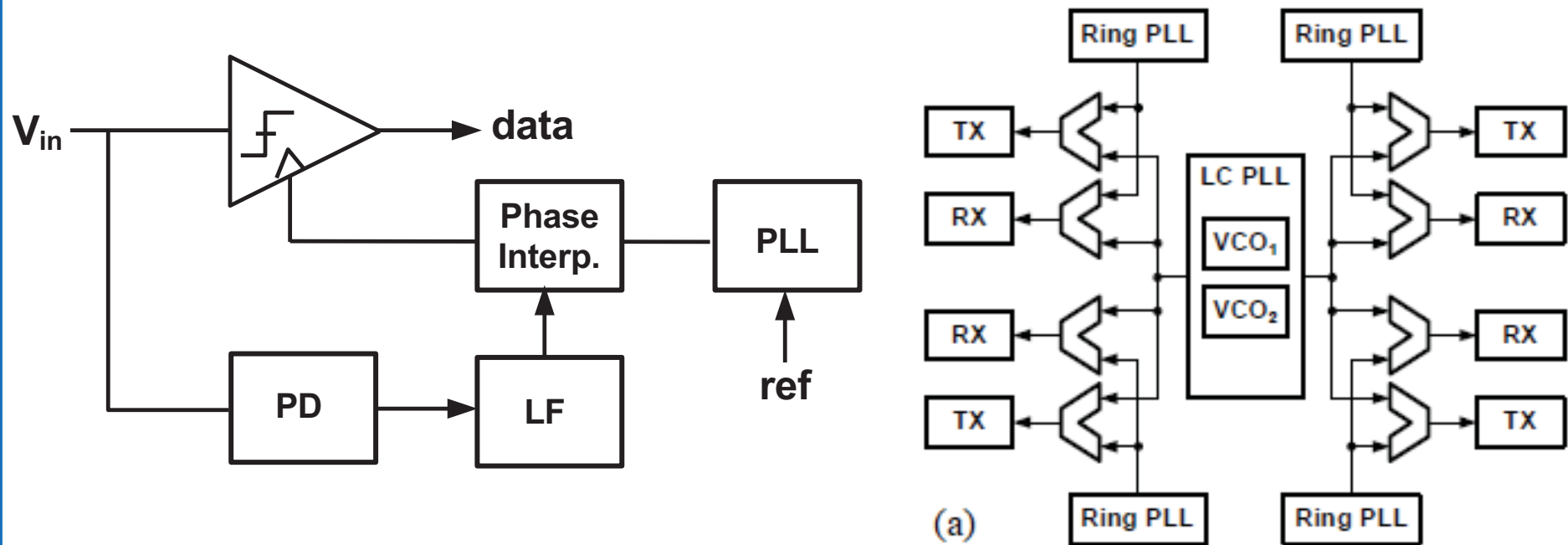
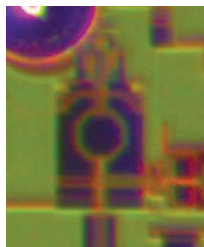
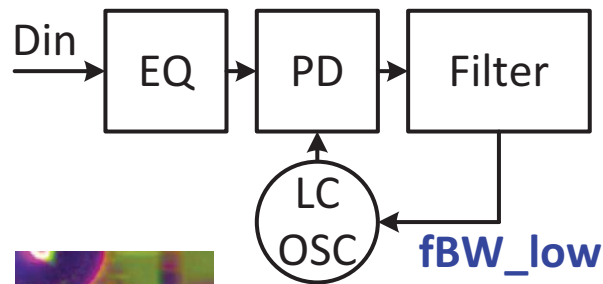
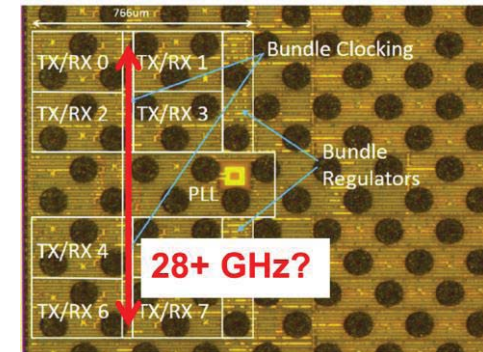


Diagram from J. Savoj et al., *CICC 2012*

- **Dual loop was standard in “golden” architecture**
 - Decouples CDR and clock generation bandwidths
 - Enables shared LC oscillators/PLLs across multiple lanes

Still Dual Loop?

- **Both motivations less clear at 56+Gb/s**
 - LC PLL doesn't need high loop bandwidth for low jitter
 - May not want to distribute 28+ GHz clock across many lanes

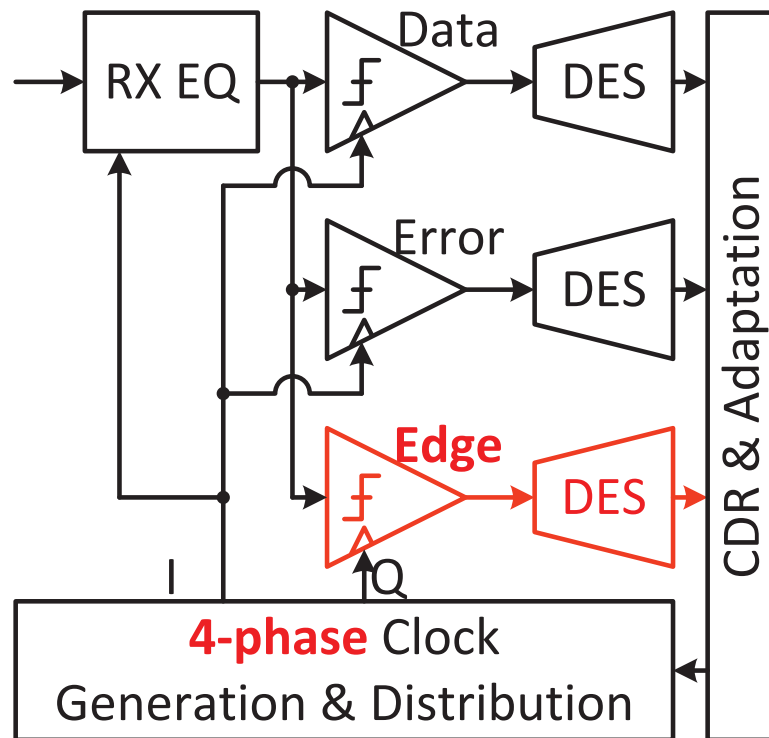


LC VCO: 0.02mm²

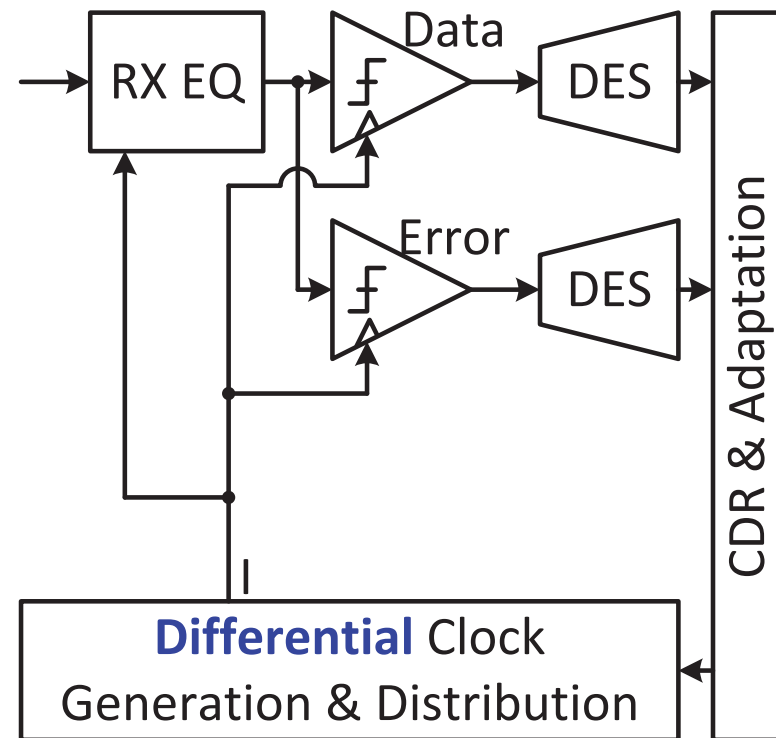
Inductor: 0.0025mm²

- **Move to single-loop design with per-lane LC PLL**
 - Inductor itself is quite compact at these rates

Oversampled vs. Baud-Rate CDR



2x oversampling CDR

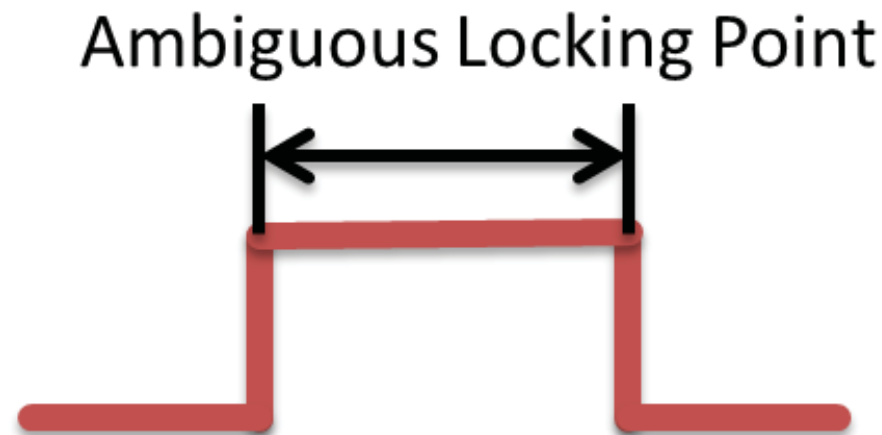


Baud-rate CDR

- **Baud-rate obviously requires less hardware**
 - Typical choice in ADC based links (2x sample rate hurts)
 - But, has some well-known limitations

Limitations of Classic Baud-Rate CDRs

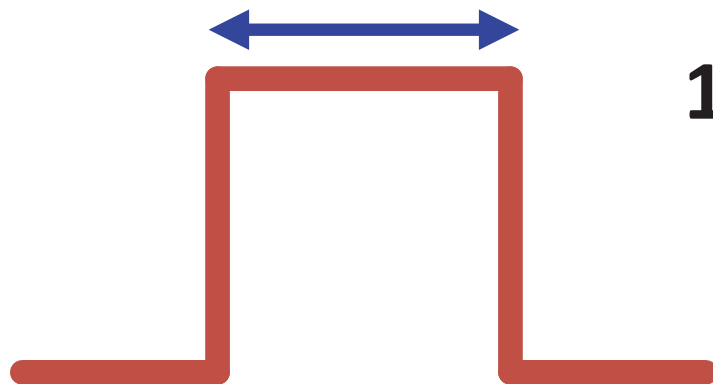
- **Mueller-Muller algorithm is most common**
 - Locks to a phase where $h[-1] = h[1]$
- **Especially in links with clean channels, there can be many phases that meet this constraint**



Avoiding Ambiguous Phase

- **Integrate-reset front-end reshapes the pulse response to have a single peak point**
 - This point corresponds to the (equalized) maximum voltage margin

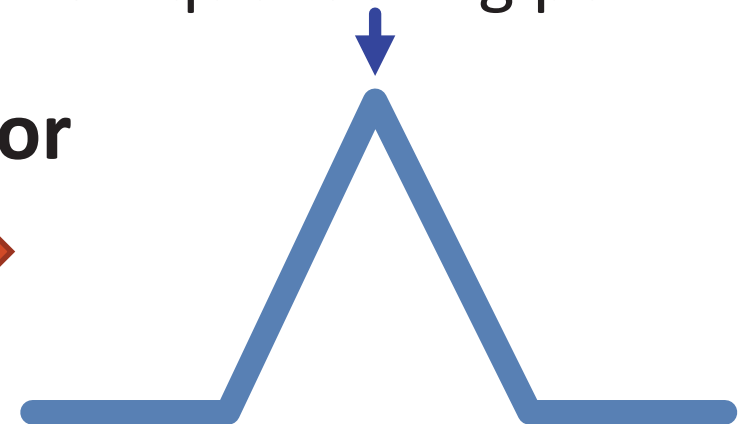
Ambiguous locking point



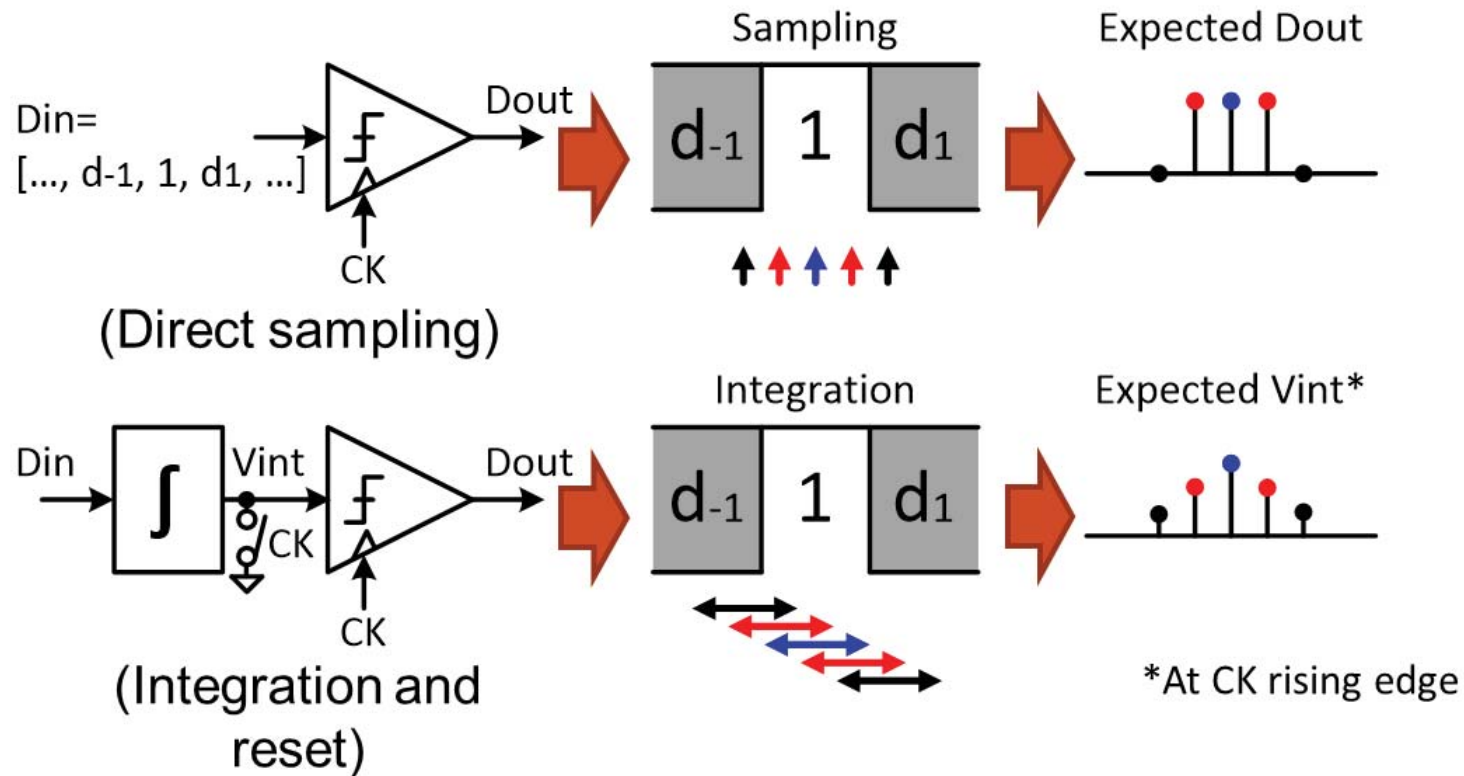
1UI Integrator



Unique locking point



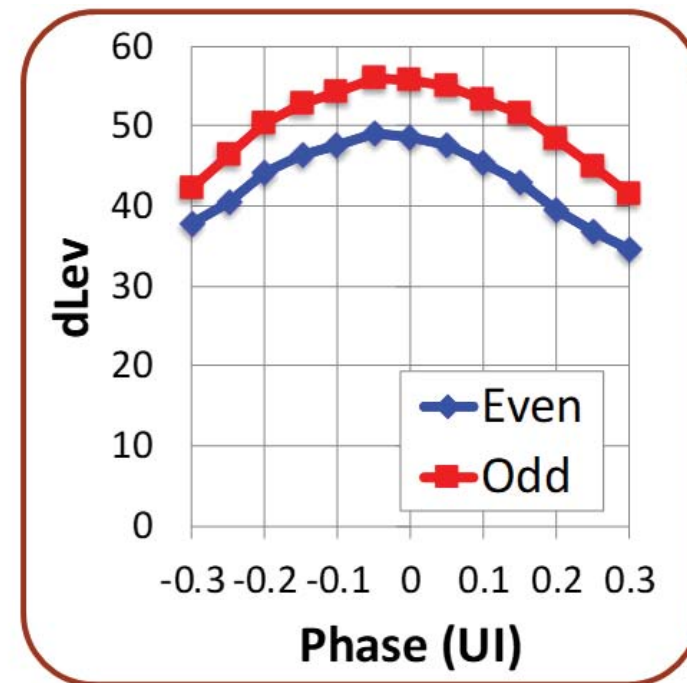
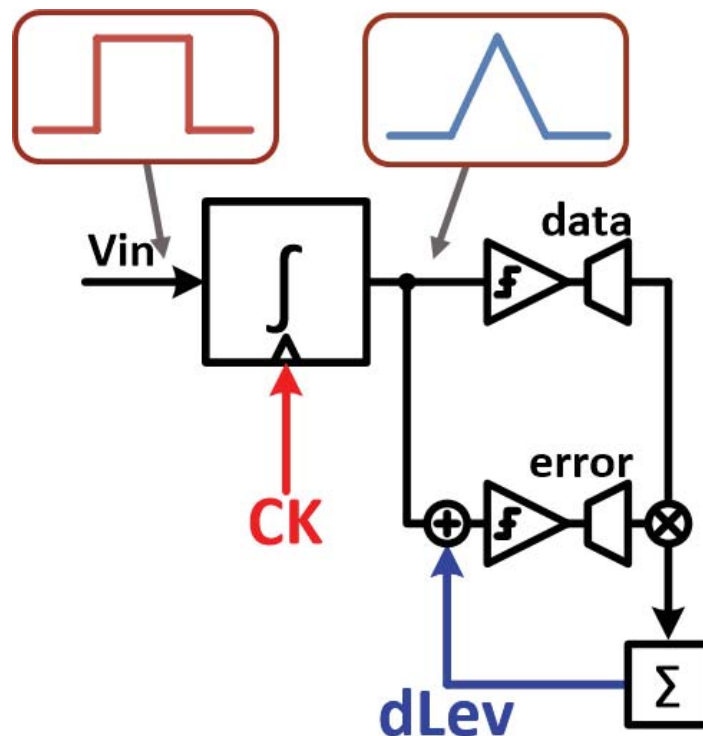
Integrating Frontend Illustration



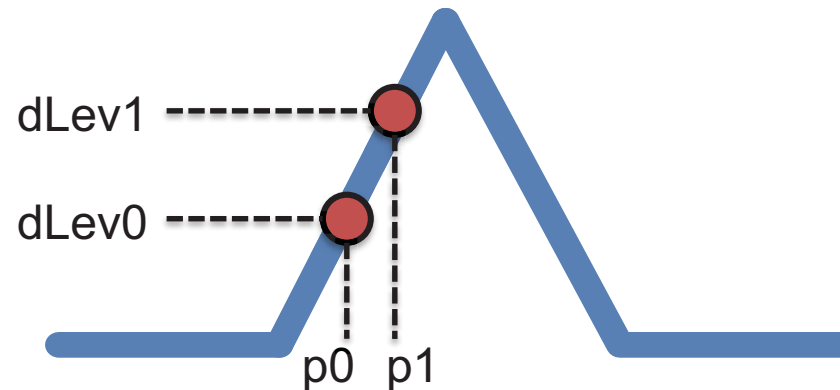
- **Re-uses CTLE (no extra hardware)**

Cursor Amplitude Estimation

- Data-level (dLev) tracking loop (for eq. adaption) re-used to estimate cursor amplitude

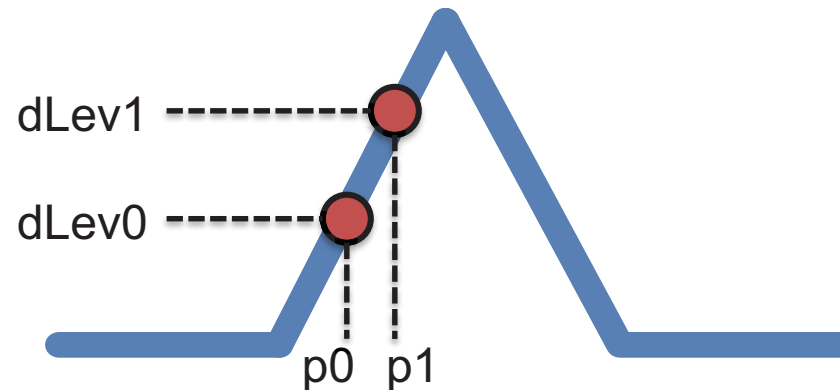


Naïve Implementation



- Adjust timing from p_0 to p_1 , measure $dLev1$
- If $dLev1 > dLev0$, stay at p_1
- Else, adjust phase in opposite direction and recheck $dLev$

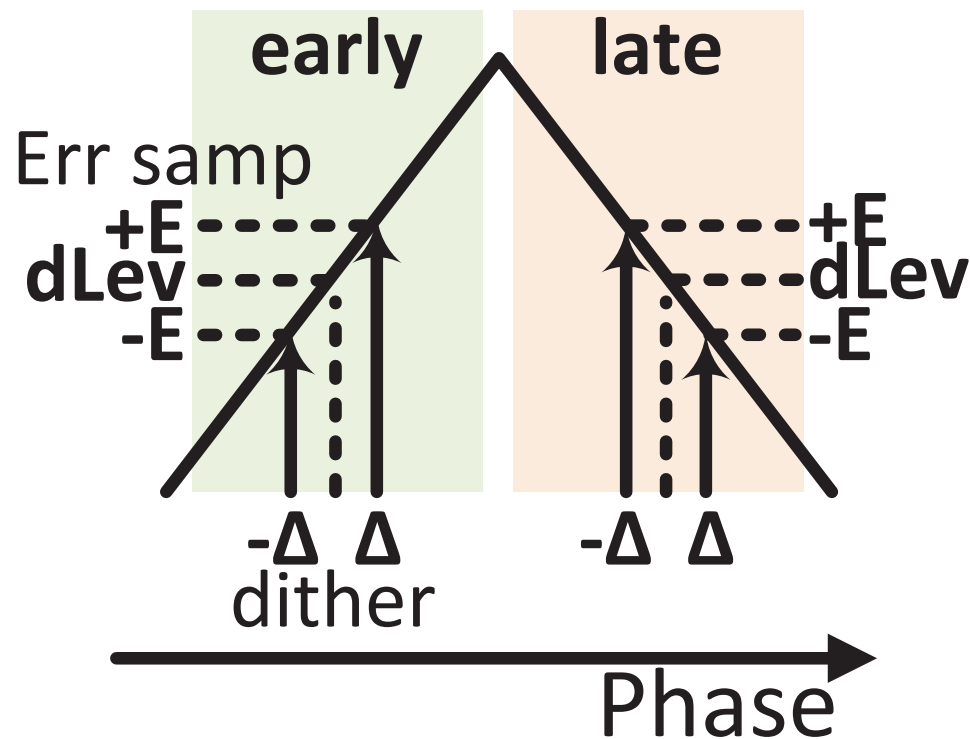
Naïve Implementation Bandwidth



- Implies that $dLev$ estimate has settled for each phase setting
- Hence, CDR bandwidth $\ll$ $dLev$ bandwidth...

Improving CDR Bandwidth

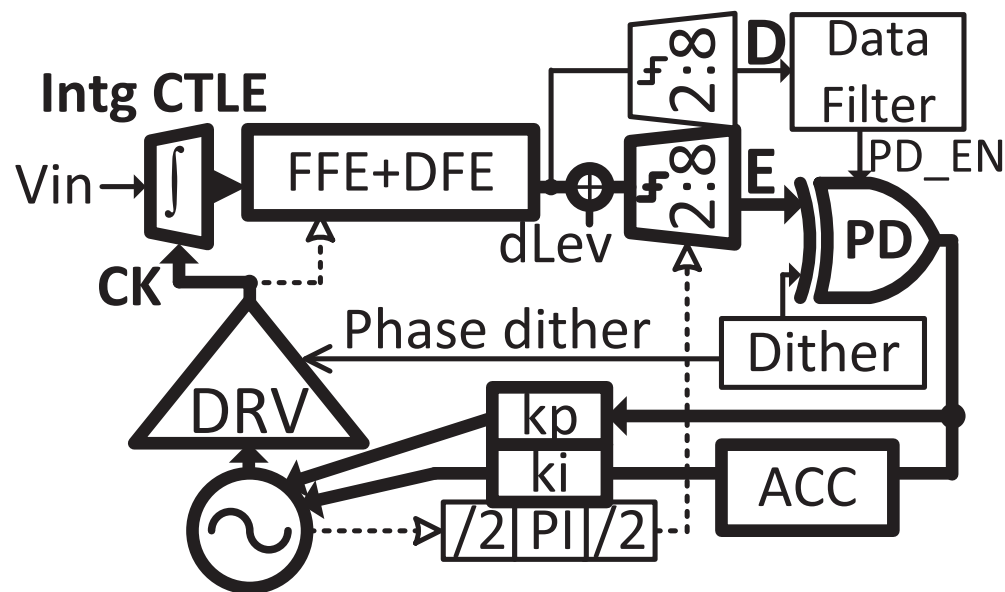
- User error sampler output instead of dLev
- Find peak by intentionally dithering phase by Δ
 - Correlation of error and Δ indicates phase error direction



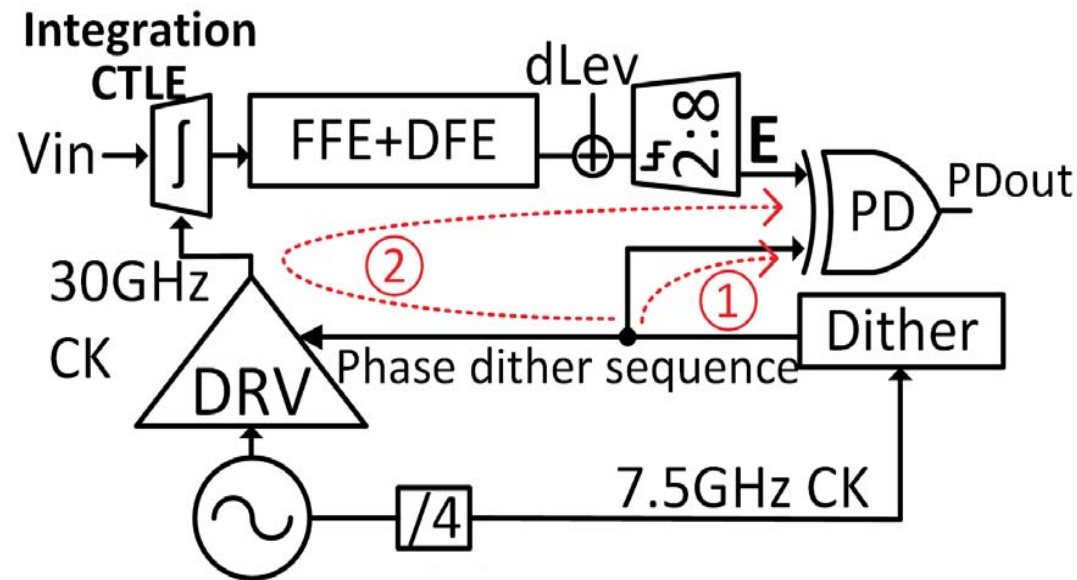
Δ	Error sampler output	PD output
0	0	Early
1	1	Early
1	0	Late
0	1	Late

CDR Architecture

- **2nd order loop**
- **Phase dither realized by clock driver delay modulation**



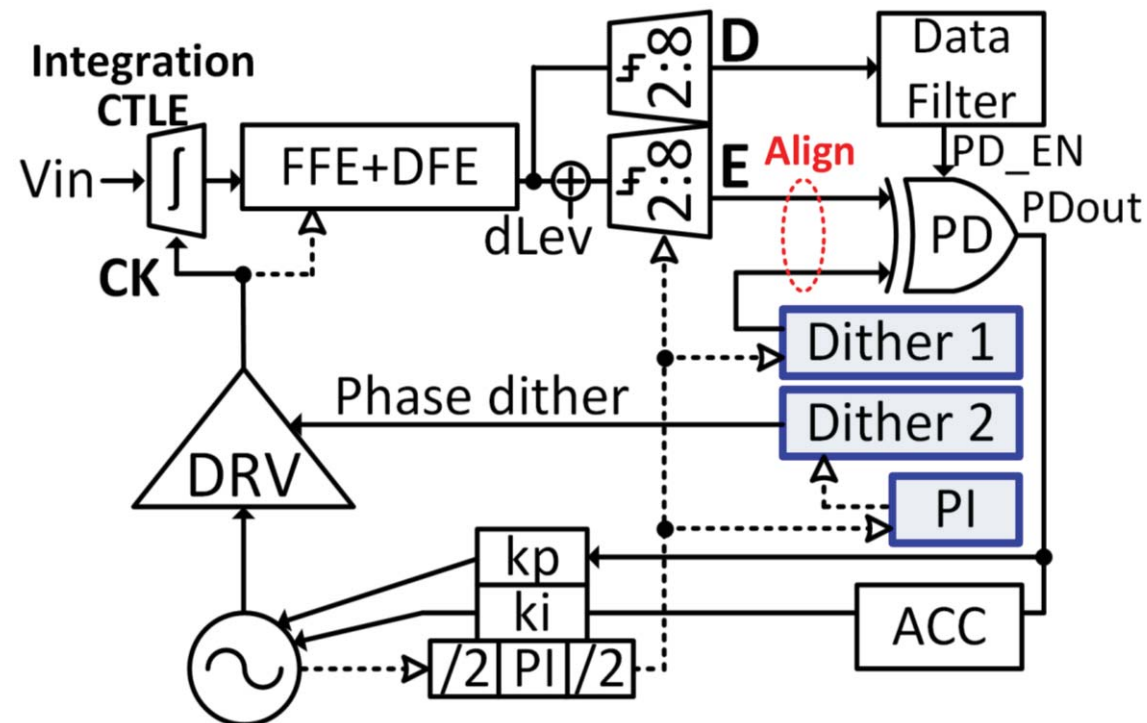
Dither Path Delay Mismatch



- **Mismatch results in PD gain loss**

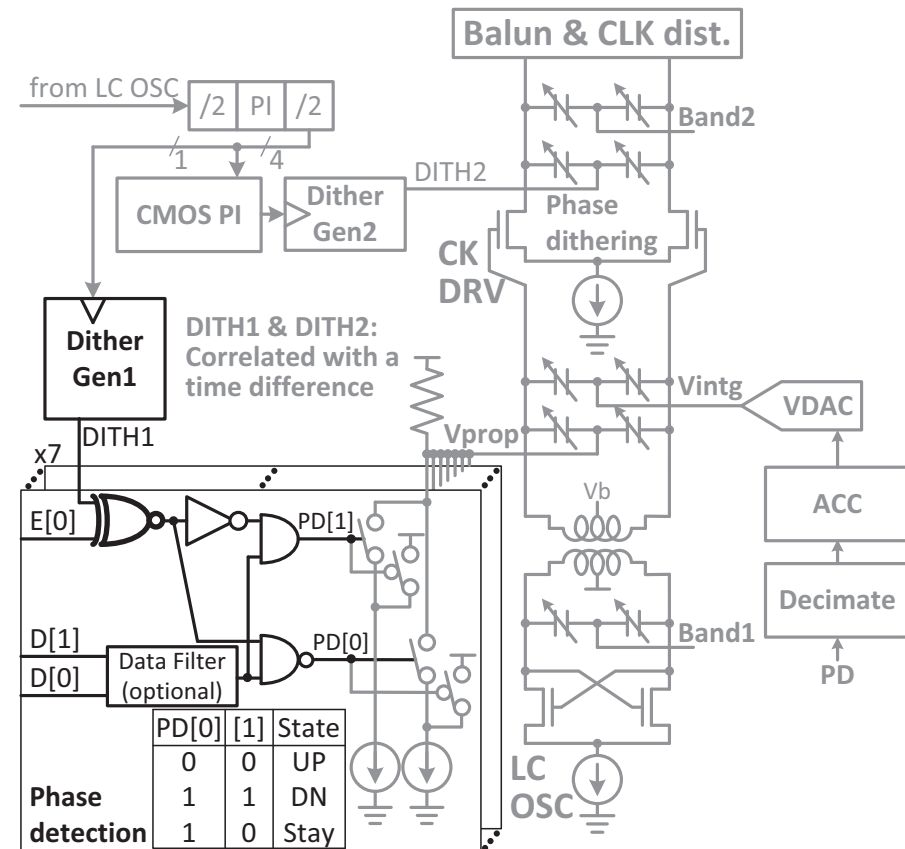
Mismatch Compensation

- Replicate dither generator with offset initial condition + phase interpolator



CDR Implementation

- LC oscillator + resonant buffer
- Dither realized by modulated buffer capacitance
- Analog summation for low-latency proportional path (@DR/8)
- Digital accumulator + DAC for integral path

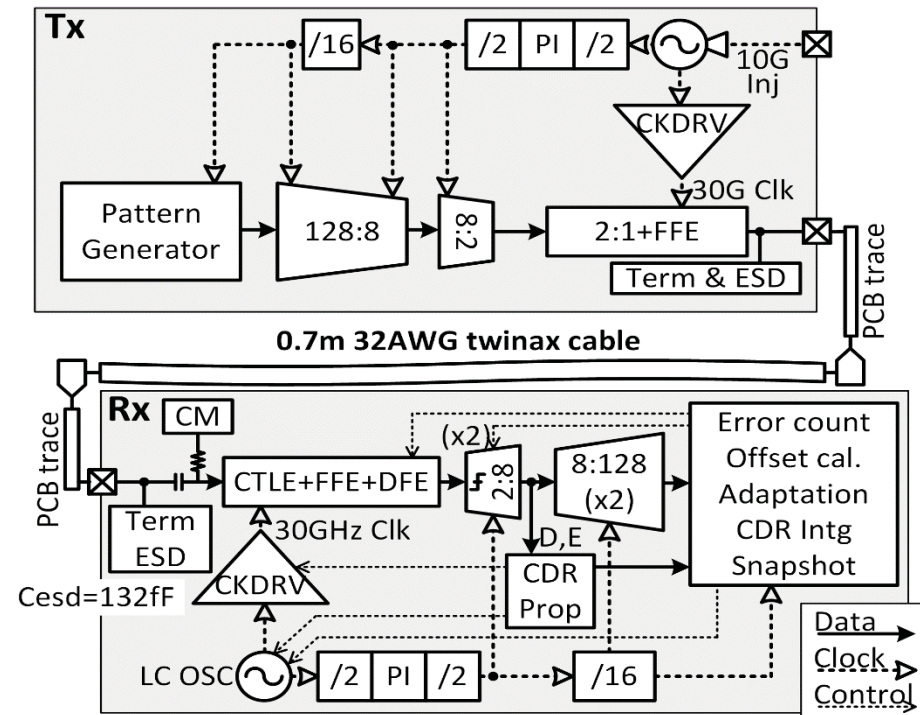


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65nm 60Gb/s NRZ Transceiver Architecture

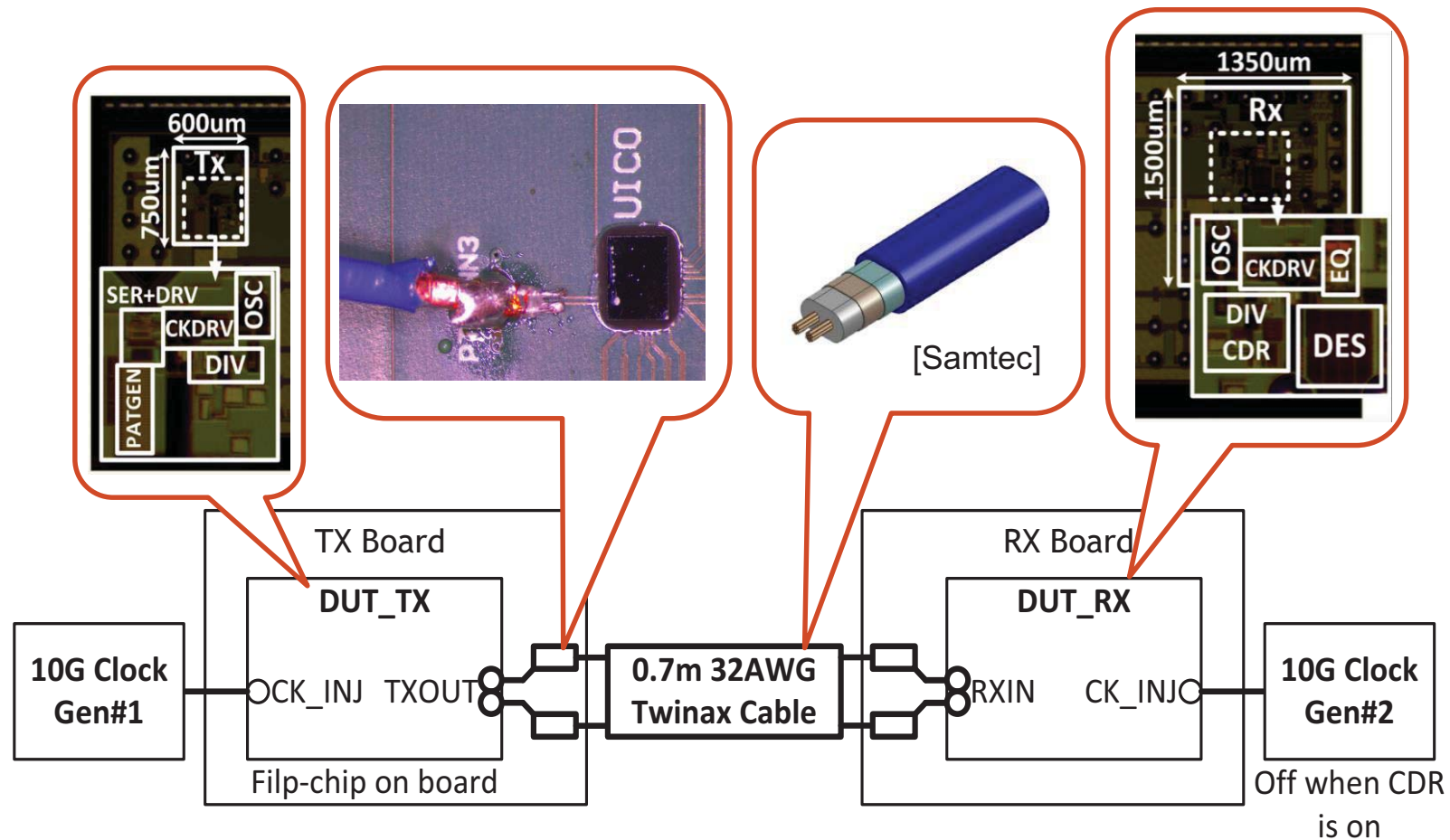
- TX: 3-tap FFE
- RX: CTLE, 2-tap FFE, 3-tap DFE
- Single-loop baud-rate CDR, equalizer adaptation engine



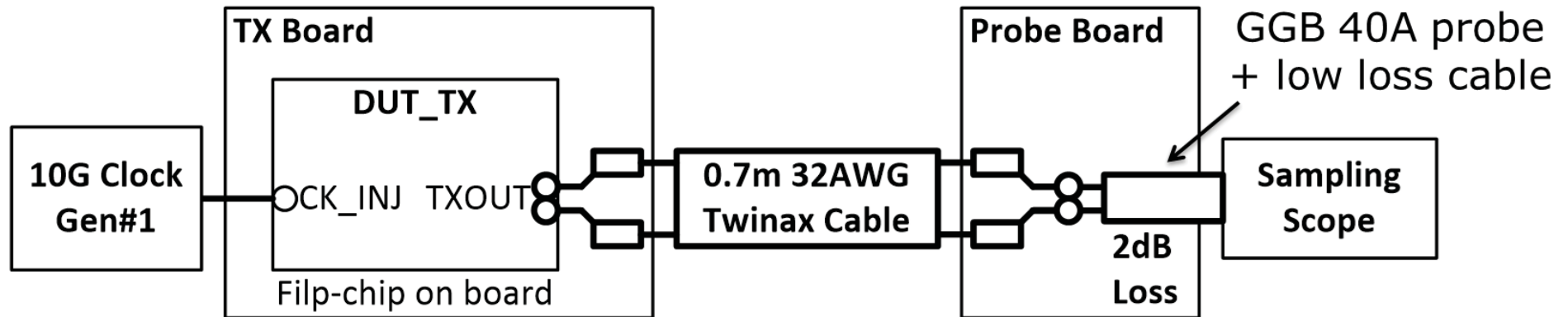
J. Han, JSSC Dec. 2017

Test Setup

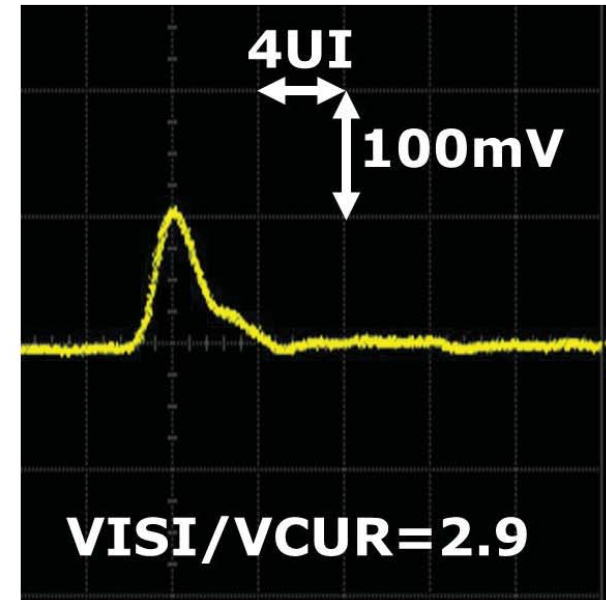
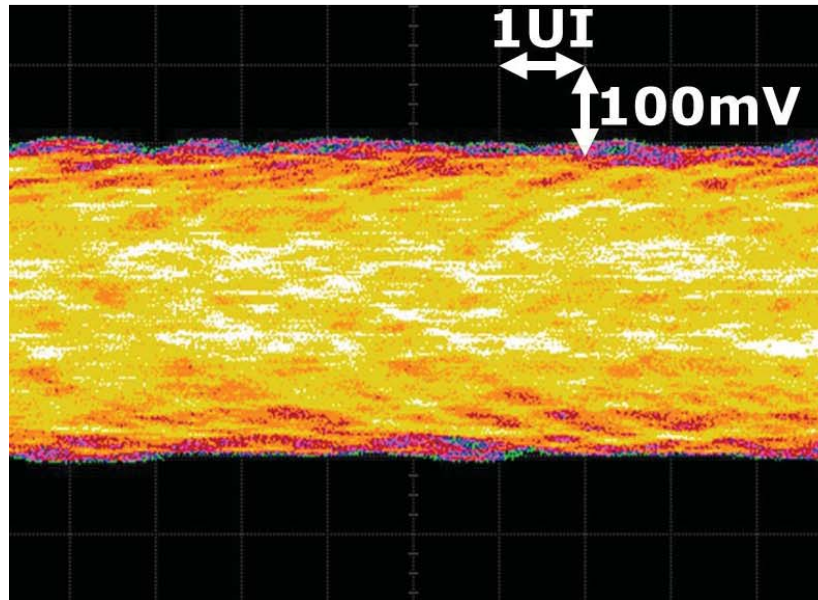
- Flip-chip on board + soldered cable



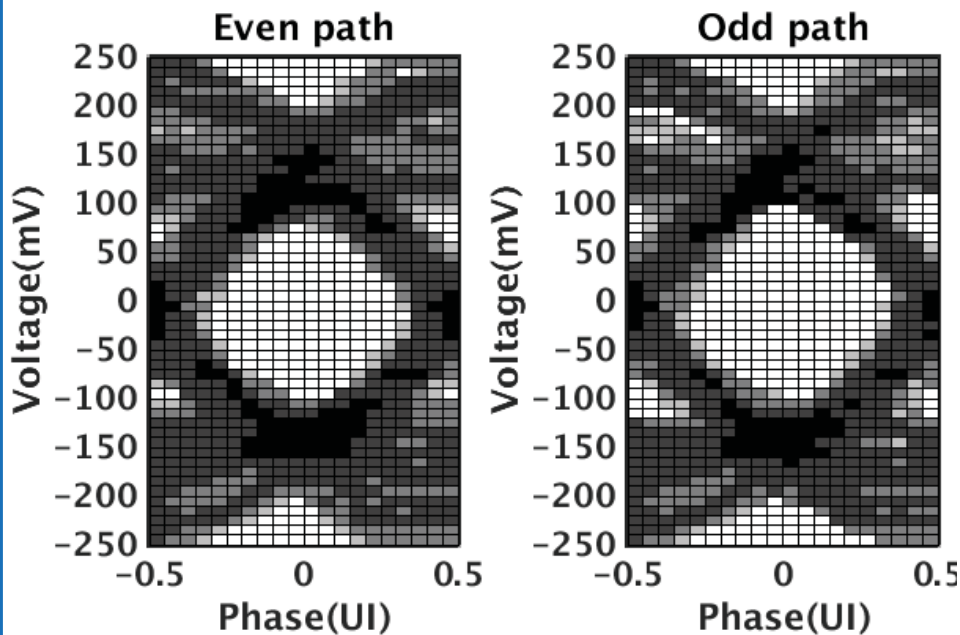
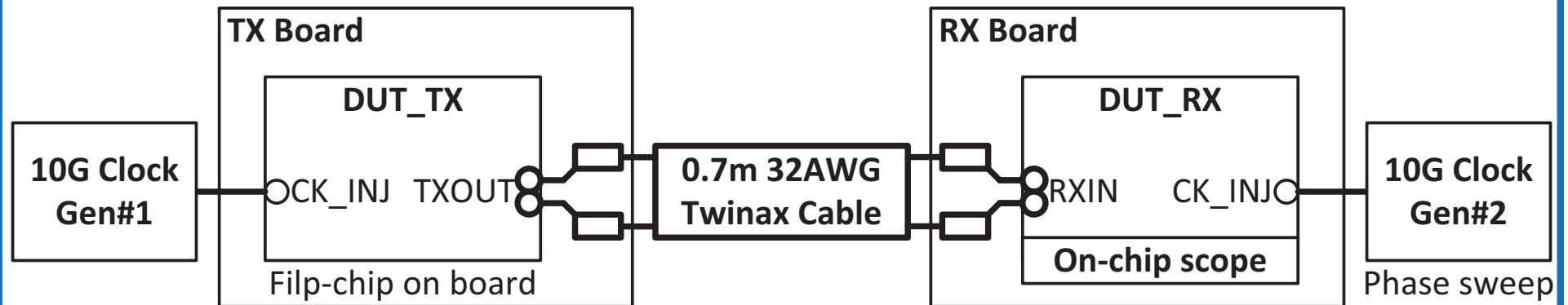
Unequalized Eye Diagram and Pulse Response



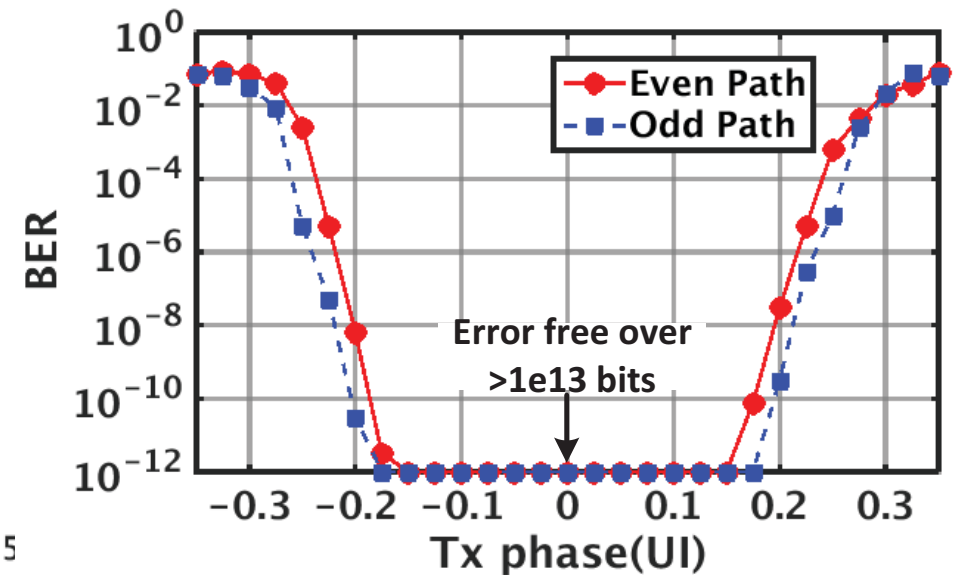
21dB
loss @
30GHz



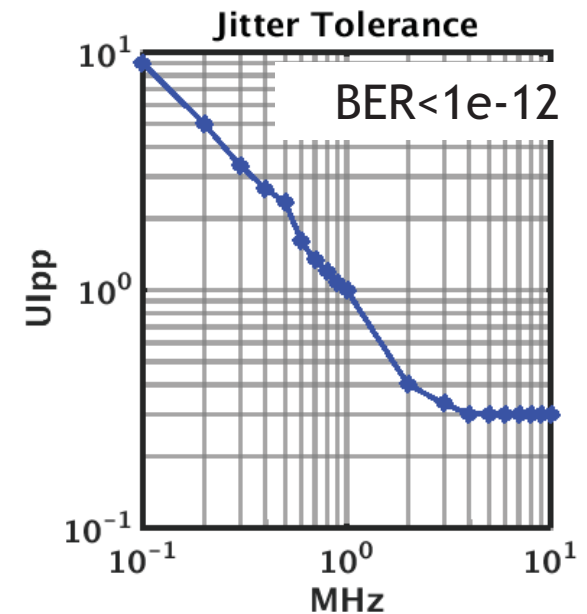
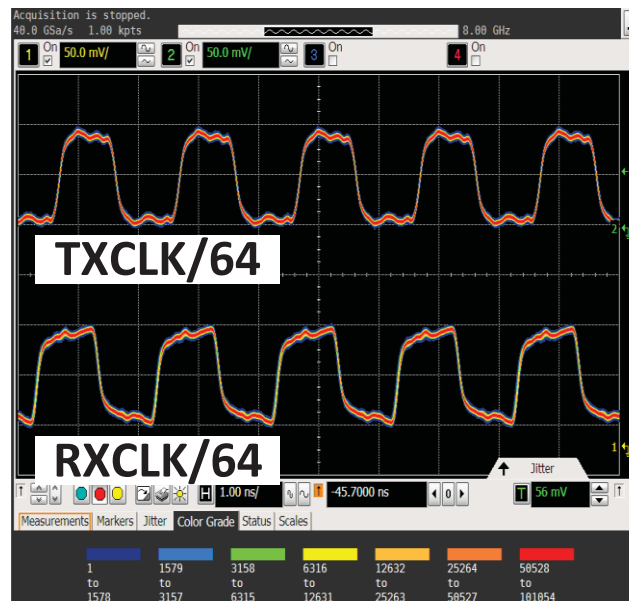
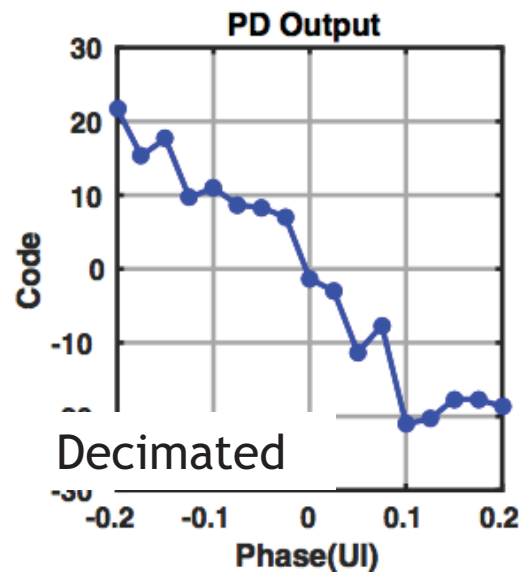
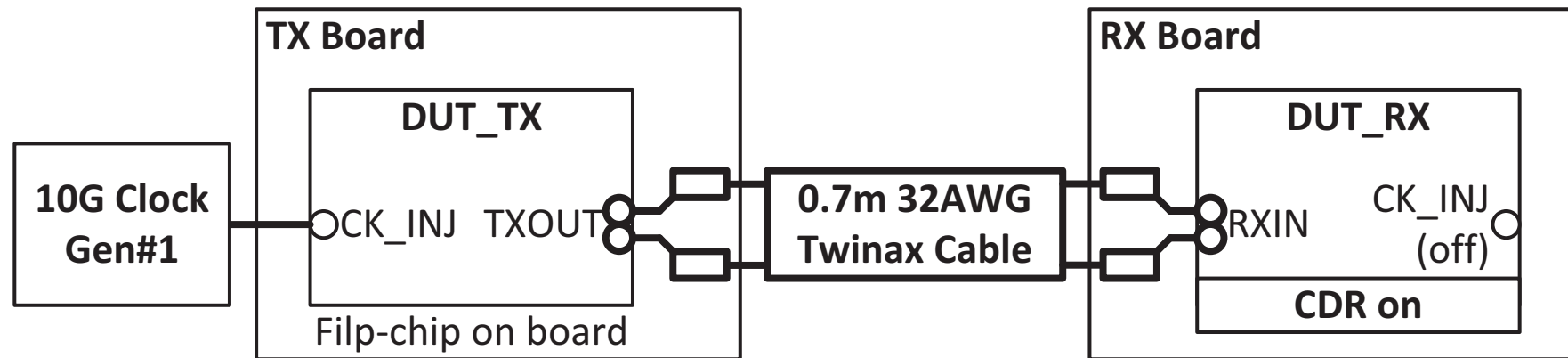
Signaling Path Performance



On-chip eyes

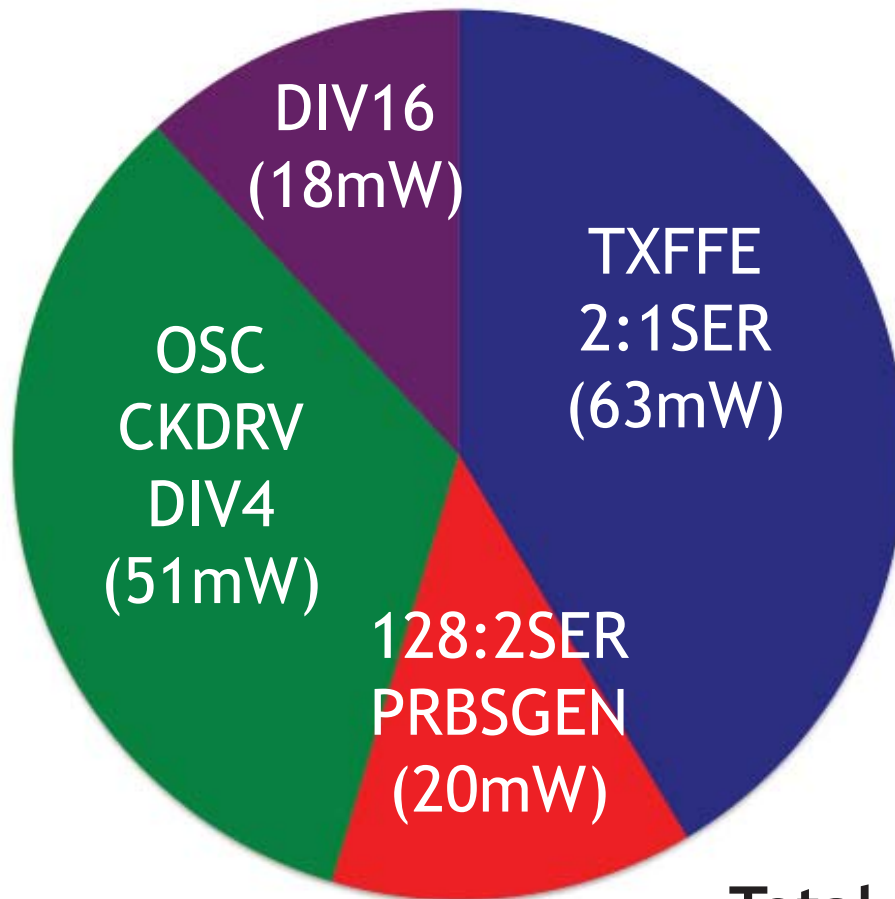


CDR Performance

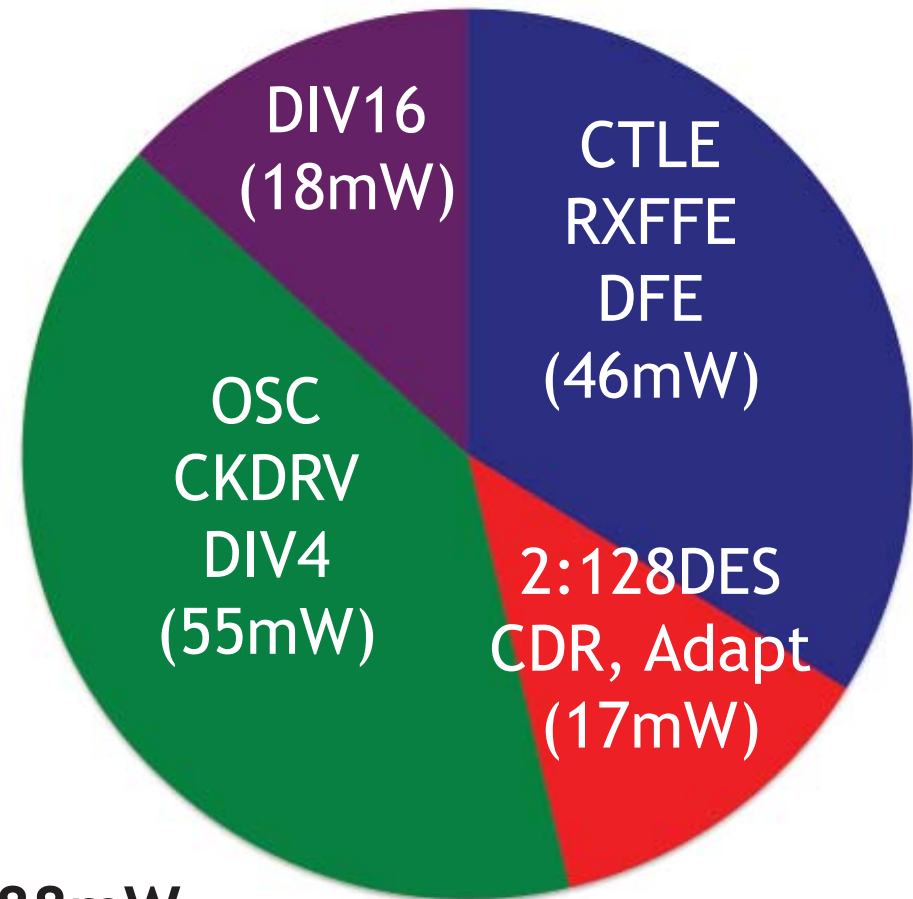


Power Breakdown

TX (152mW)



RX (136mW)



Total 288mW

Outline

- Introduction
- Power in SerDes Circuits
- Implications on SerDes Architectures
- 60Gb/s 65nm Design Example
- **Conclusions**

Conclusions

- Increasing link rates to 56+Gb/s requires improved energy-efficiency
- Analog circuits “win” over digital when dynamic range is relatively low
 - Main source of dynamic range in links is channel loss – 40dB is only ~6.5 bits
 - Mixed-signal techniques will therefore remain critical
- **Keys to mixed-signal circuit efficiency:**
 - Stay away from intrinsic speed limit (current integration, parallelism, gain outside DFE loop, ...)
 - Keep dynamic range off of the signal path (eq. architecture, baud rate, cascode gain control, ...)